

Faculty of Engineering
Computer department
Graduation Project 2021

Project Title

**Managing Of Quarantine
System**

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Breakdown of individual contributions

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1. Mobile application for patient.
2. Backend programming with flask.

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1. Application for Doctor and caravan.
2. Allocation and distribution algorithm.

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1. Implement chat function.
2. Hospital desktop application.

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1. Build Machine Learning Predictive Model with Logistic Regression.

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1. IOT monitoring and measuring devices.

Abstract

Managing of quarantine Mobile application allows Cities and National Crisis Service Organizations to offer the public with a comprehensive personal monitoring app. The Managing of quarantine System allows municipal and national health organizations to enhance the public's sense of safety, wellbeing and interaction with response services, and at the same time allows the national health organization to monitor people who are in quarantine and need to be isolated, while raising the government's trust in the eye of the public

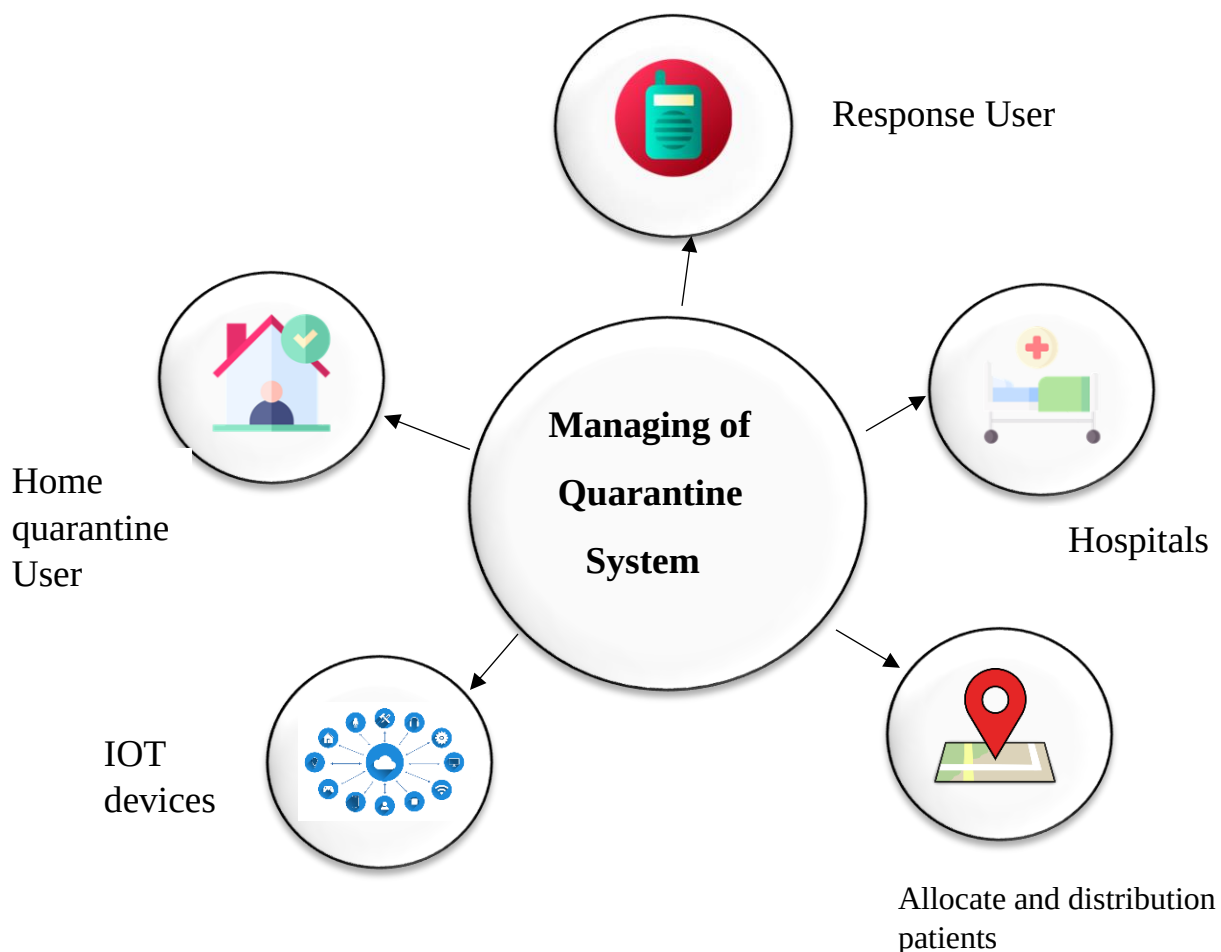


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Chapter 1: Introduction

Last 2 years our world face very large epidemic called “covid-19. Last 2 years our world face very large epidemic called “covid-19. By March 2021, 2.86 million persons world-wide had died from COVID-19 infection, 132 million had tested positive. Overall case-fatality rate is very high for COVID-19 infection and depending on the phase of the epidemic, testing, registration, and demography, healthcare capacity and governmental decisions.

This virus causes very large damages in all world countries’ economy. Its spread has left national economies and businesses counting the costs, as governments struggle with new lockdown measures to tackle the spread of the virus.

Most people who fall sick with COVID-19 will experience mild to moderate symptoms and recover without special treatment.

The virus that causes COVID-19 is mainly transmitted through droplets generated when an infected person coughs, sneezes, or exhales. These droplets are too heavy to hang in the air, and quickly fall on floors or surfaces.

You can be infected by breathing in the virus if you are within close proximity of someone who has COVID-19, or by touching a contaminated surface and then your eyes, nose or mouth.

This virus causes very large damages in all world countries’ economy.

All these things lead us to think about some effective solution that help infected people and reduce infection.

1.1 Problem statement

Transmission of infectious diseases is often prevented by quarantine and isolation of the populations at risk. These approaches restrict the mobility, social interactions, and daily activities of the affected individuals.

Quarantine and isolation are public health measures used to prevent the transmission of infectious diseases among individuals and communities.

It is difficult to manage and organize the quarantine without system to do this, because some patients with moderate infection need to quarantine at home and others with severe infection must quarantine at hospital.

1.2 Objectives

Implement a system consist of (Smartphone applications based on machine learning enabled with IOT) using information such as Global Positioning System (GPS) and Geographic Information System (GIS) for tracking purposes have been widely used during the COVID-19 pandemic in order to increase the chance of detecting infected people and IOT devices to monitoring them.

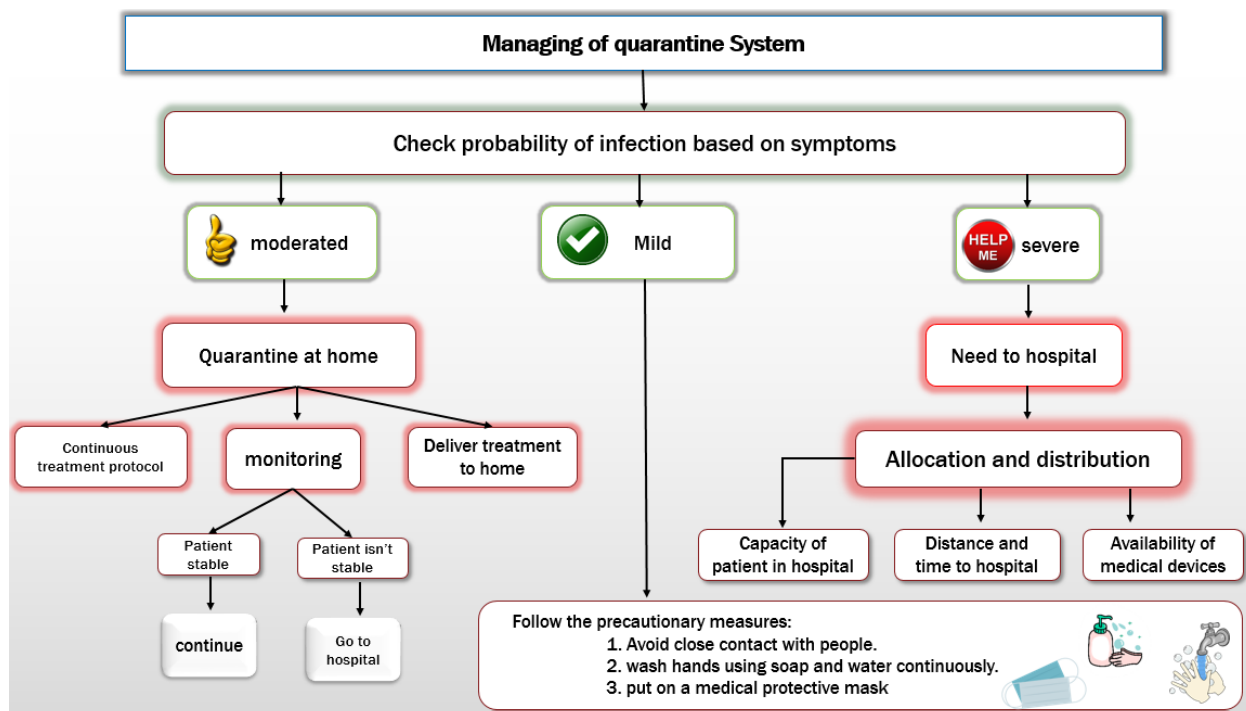


Figure 0-1 (Project description)

2 Chapter 2: similar projects

2.1 Patient Telemonitoring System Using Biomedical Sensor Network

In this project, they have designed IoT based remote patient monitoring system using biomedical sensor network and GSM alarm triggering system for monitoring the Oxygen saturation level, temperature, heart rate, blood pressure and measuring the electronic signals generated by heart pulses of the patient's body. This proposed system can draw a significant role to improve the e-healthcare system. A GSM module have been used in this system so that whenever any particular health parameter goes beyond the ideal limit or any abnormal situation arises, the SMS alarm system would send text SMS, mentioning relevant cause, to a pre-defined mobile number. Through this SMS, the corresponding doctor or nurse would be notified instantly allowing to provide the proper treatment in time. In this IoT system, the data are collected through the Arduino Mega 2560 and are stored in the dedicated ThingSpeak channel through the Wi-Fi internet system. This allows the concerned doctor to access data of patient's real-time condition from anywhere in the world through the user interface webpage. This system is introduced so that doctors, nurses or any other health worker can monitor the real-time condition of patient from anywhere in the world without any barrier. Anyone wanting to monitor the data of patient's condition must login with channel identity number and password dedicated for the specific channel. There is also an emergency button which allows patients to call the doctor, nurse or relatives in any emergency situation. The system will basically accelerate the agility of patient monitoring process by applying telemonitoring method mostly for hospitals by connecting every process in one single system in real-time frame [18]. This undoubtedly reduce discontinuations and delays caused by physical distance and complex environment. In fact, the proposed system would contribute to make the emergency response of a medical system more efficient.

2.2 Corona-Warn-App

The Robert Koch Institute (RKI) – in its capacity as the central federal institution for public health and as the German national public health institute – publishes the Corona-Warn-App on behalf of the German government. The app serves as a digital complement to distancing,

hygiene and wearing masks. Additionally, it provides a functionality to add your digital vaccination certificates to prove your vaccination status. It uses Bluetooth technology and the Google Exposure Notification APIs. The app is designed to help break infection chains by informing you if you were recently close to someone who subsequently tested positive for corona. However, at no time does it collect any personal information. Who you are and where you are remain secret – and your privacy is well-protected.

HOW THE APP WORKS

Exposure logging is the core of the app and should always be active. When enabled, smartphones exchange encrypted random IDs with other devices using Bluetooth.

The random IDs only provide information about the duration and distance of an encounter. No one is able to identify the person behind these IDs. The Corona-Warn-App does not collect any information about the location of the encounter or that of its users.

Based on maximum corona incubation times, the random IDs your smartphone collects are stored in an exposure log for 14 days and then deleted. If a person is diagnosed with corona and chooses to share his/her own smartphone's random IDs, all persons previously encountered will then be notified anonymously.

No one will know when, where or with whom the exposure event took place. The infected person remains anonymous.

In addition to the exposure notifications, affected users will receive clear recommendations for action. Important: Information about the notified persons is not accessible at any time.

3 Chapter 3: Mobile Application Phases



3.1 Patient's mobile application

3.1.1 Overview

- Covid online test:

The person who do this test is asked many question by the system and depend on this questions system defines its case (mild infection, moderate infection and severe infection).

1. If patient case is severe infection that means that this patient need to move to hospital.
2. If patient case is moderate system send test result to doctor and start home quarantine.

- hospital allocation

In this part if system show that patient need hospital it choose the best hospital for patient depending on his location and his case.

Ex: if patient suffers from heart problems he must move to hospital that contains heart department.

- Allocate patients to suitable doctors if patient don't need to move hospital.
- Display information about corona virus and how to interact with infected people.
- Treatment protocol for patient during quarantine with each medicament time and potion.

3.1.2 Flow chart of the system

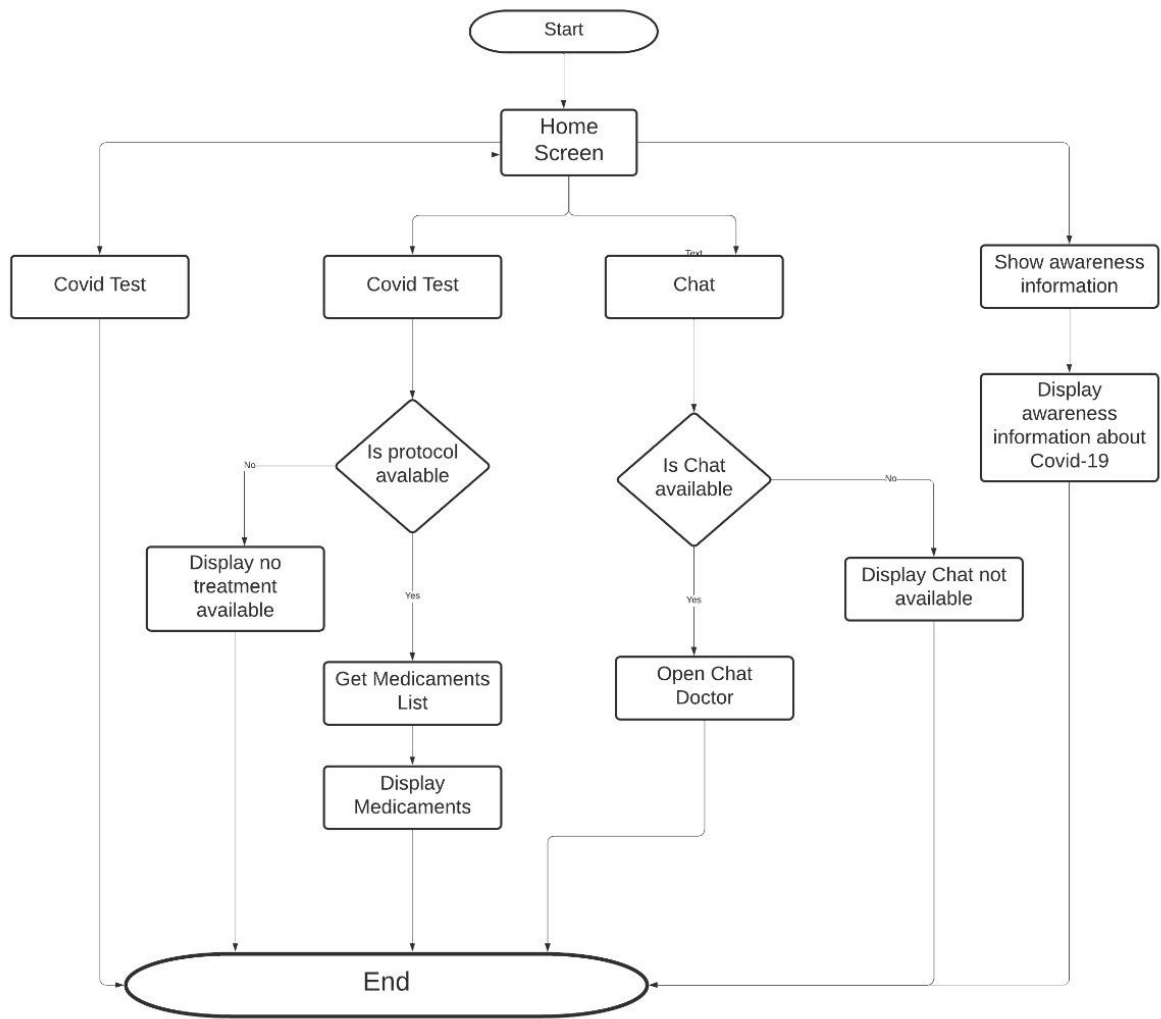


Figure 2patient app flow chart

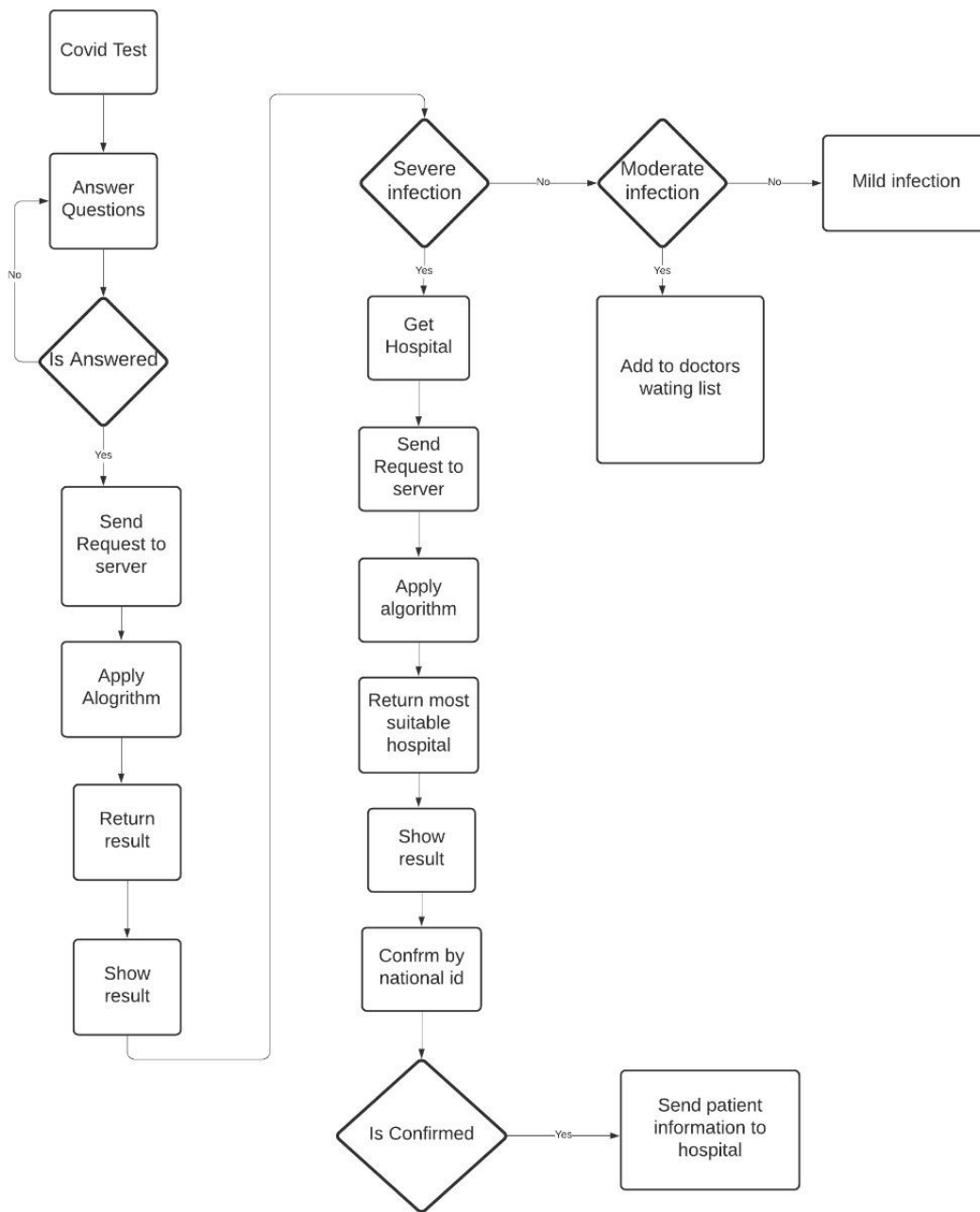


Figure 3covid test

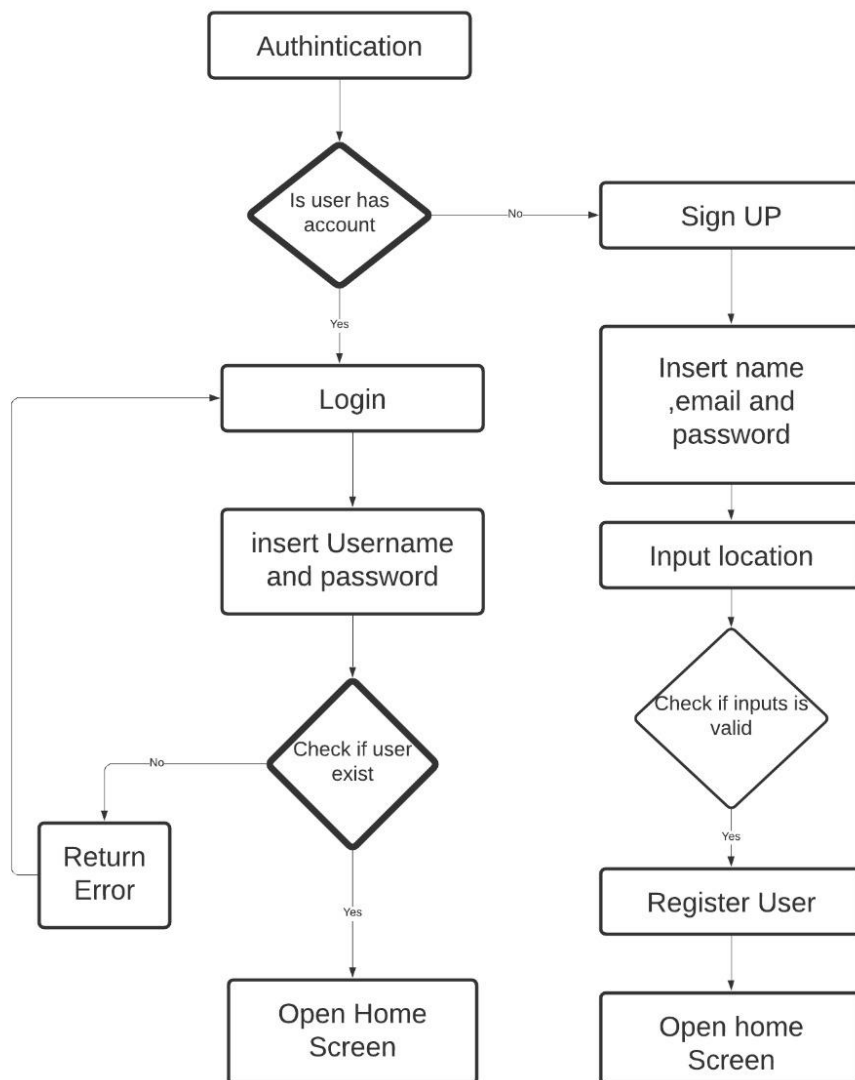


Figure 4patient authentication

3.1.3 Application description

3.1.3.1 Login screen

- If user has account in the system he will login by username and password.
- The application will send this data to database and check if this inputs is correct and user is exist in the system .
- If user is registered in the system he will directed to the home page of the application.If not registered error message will be appeared tell him that username and password is not correct.
- If user don't have account ha can create new account and use the system by pressing sign up.

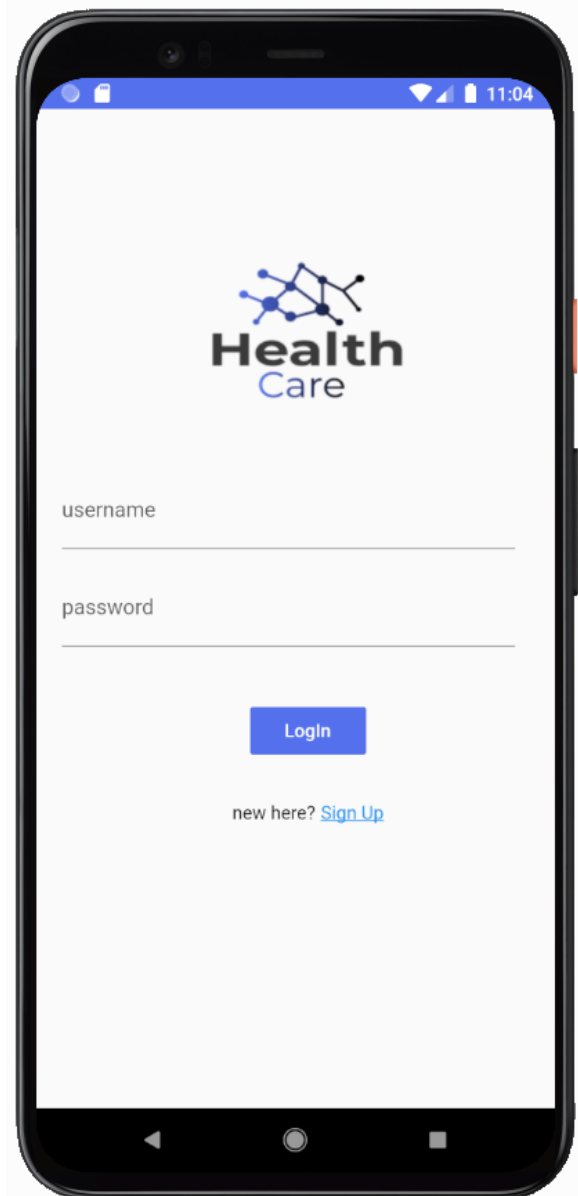


Figure 5login

3.1.3.2 Registration screen

- This screen is responsible for creating accounts for new users in the system.
- Application asks the user to enter his name, username and choose secure password.
- User must enter his location by press the button of location and open GPS on his mobile the location will be added automatically.
- Application will check if this data is correct and post it to backend to check if the username is exist in system database or not .
- If username is exist application gives error to add another username.
- If inputs is true and user not exist in database system will register the new user and open home screen.

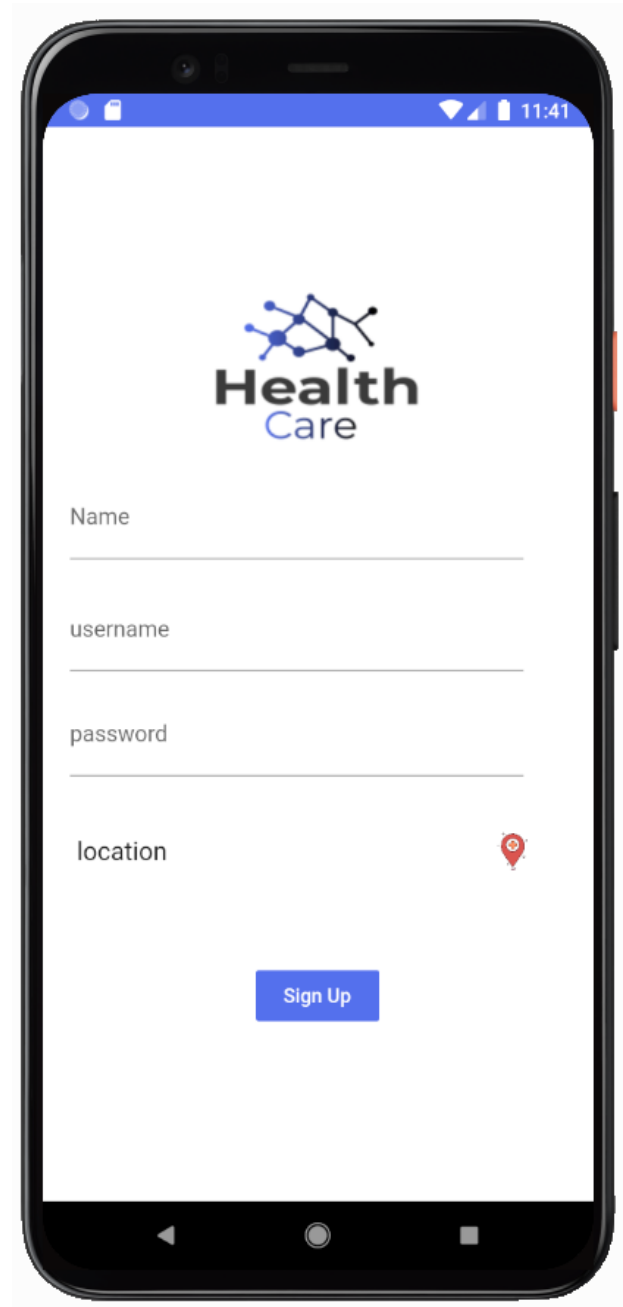


Figure 6registration

3.1.3.3 Home screen

- After user signed in he will directed to home screen of the application.
- Many choices will be available for him :
 - 1- Making simple test to check if he is infected with covid-19 or not.
 - 2- If patient in quarantine he will be able to see its treatment protocol and chat with the doctor who is responsible for his case.
 - 3- He can see awareness information about covid-19 and how can he take care about himself.

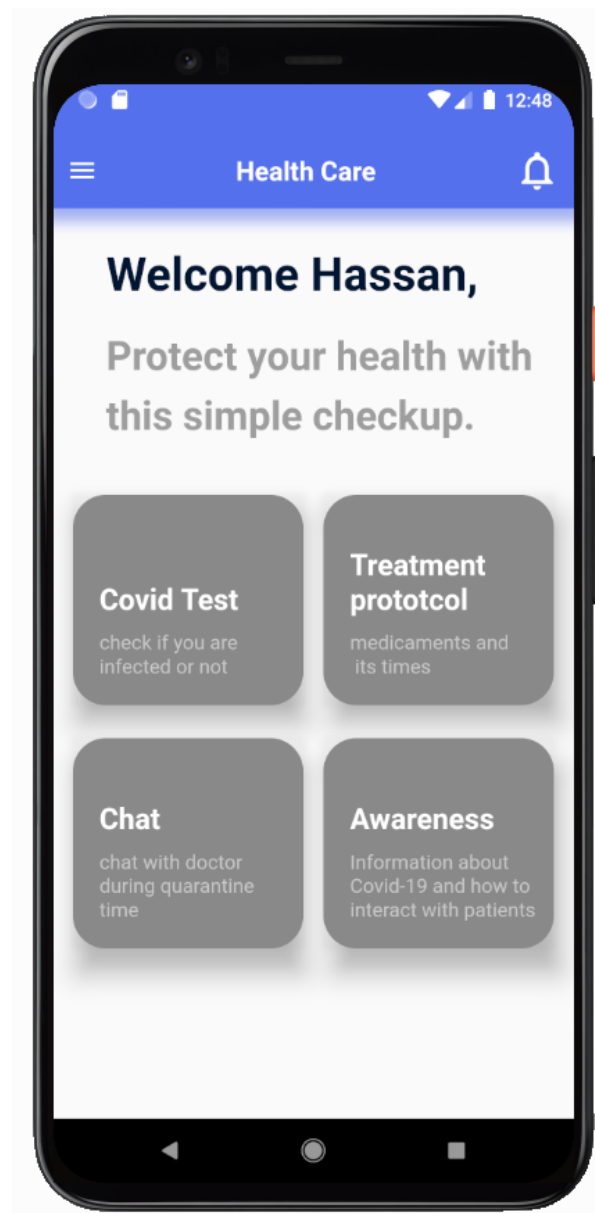


Figure 7mobile app home

3.1.3.4 Covid-19 test

- Covid test is a simple test for corona virus that checks if user is infected with covid-19 or not and what is the degree of infection.
- First user open the test it asked him to answer 9 questions this questions contains information about age, if he feels tiredness or not, body pain, dry cough, if the patient feels difficulty in breathing, if this patient contact with infected person.
- This questions is required to determine the degree of infection.
- After answering questions the result is sent to the server which is built by python framework (Flask).
- Server receive the data and apply machine learning algorithm.
- Then return the result of this process.
- Result contains the degree of infection of this patient.

The screenshot shows a mobile application interface for a 'Covid Test'. The app has a blue header with 'Health Care' and a notification bell icon. The title 'Covid Test' is displayed in blue. The form contains the following fields and their selected values:

Question	Selected Answer
Age	
Gender	male
Tiredness & bodypain	no pain
dry cough	no
difficulty breathing	no difficulty
runny nose	no
nasal congestion	no
Fever effect	
contact infected person?	don't know

A blue button labeled 'Result' is located at the bottom of the form.

Figure 8covid-test

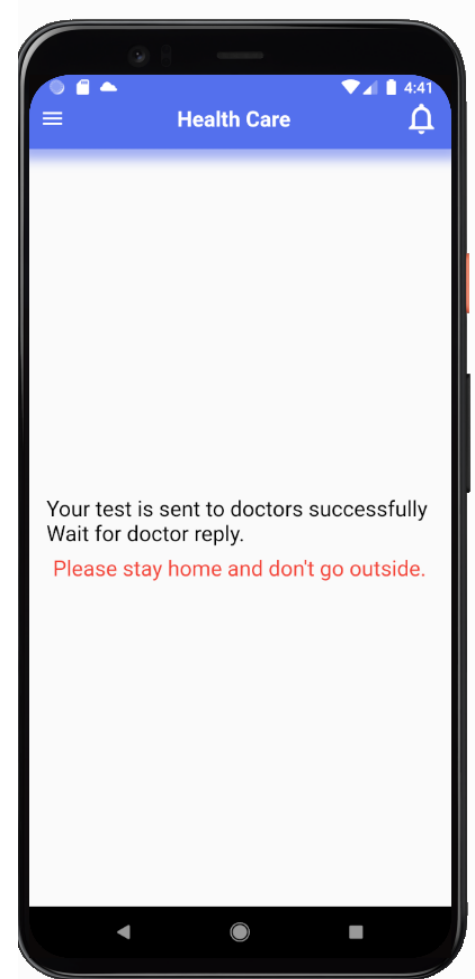
3.1.3.5 Result

- After sending data to server and make processing on it the result is returned by to mobile application.
- This result contains the degree of infection and patient case.
- Now we have 3 cases algorithm will return one of them:
1- severe infection

- The most dangerous case that infected person has difficulty in breathing.
- In this case he need to move to hospital as fast as possible.
- Application takes his location and apply algorithm to get the most suitable hospital for his case.
- The input parameters for this algorithm is the location of the patient and degree of infection.
- This algorithms return suitable hospital based on distance between this hospital and patient, available devices and free beds in hospital.

2- moderate infection

- In this case patient don't need hospital just make a home quarantine and communicate with doctor.
- Patient data is uploaded to database and doctor will see it in his application.
- Doctor will add a prescription after see patient data this prescription will appear in the application.
- If required doctor can open chat with patient.
- After successfully sending data patient is asked to stay home and don't go outside until doctor see his data.



3- Mild infection

In this case patient don't need neither a hospital nor home quarantine just stay at home until syndromes disappear.

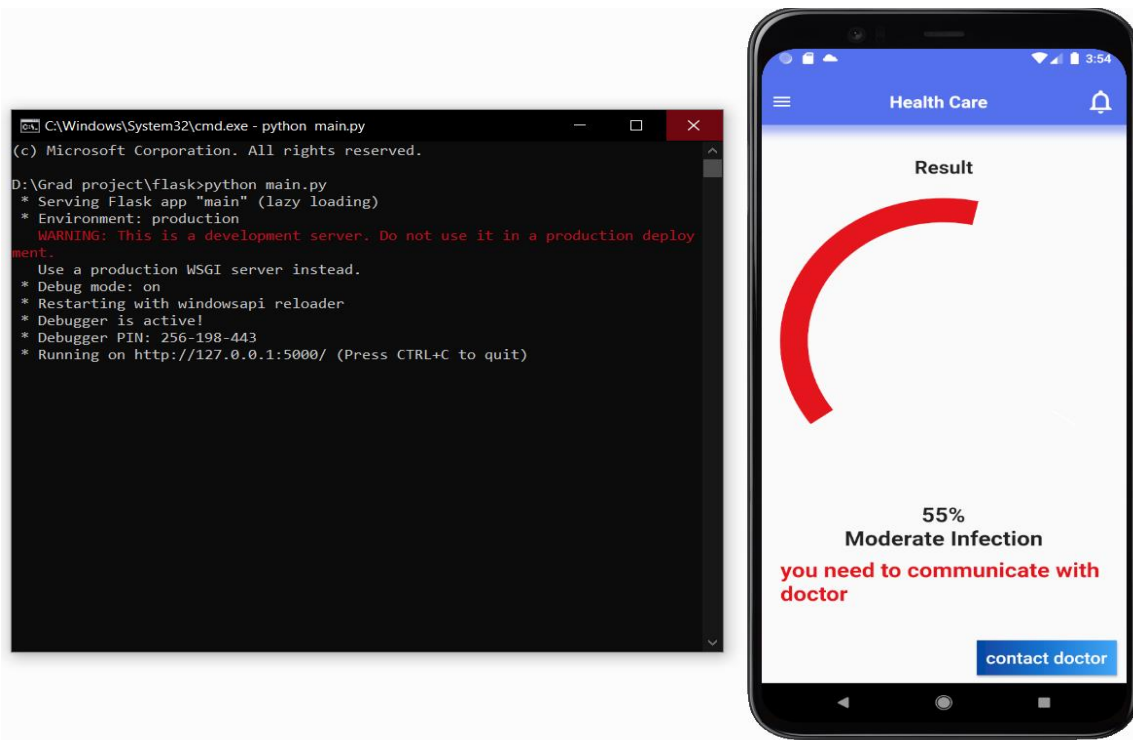


Figure 9 connect to server

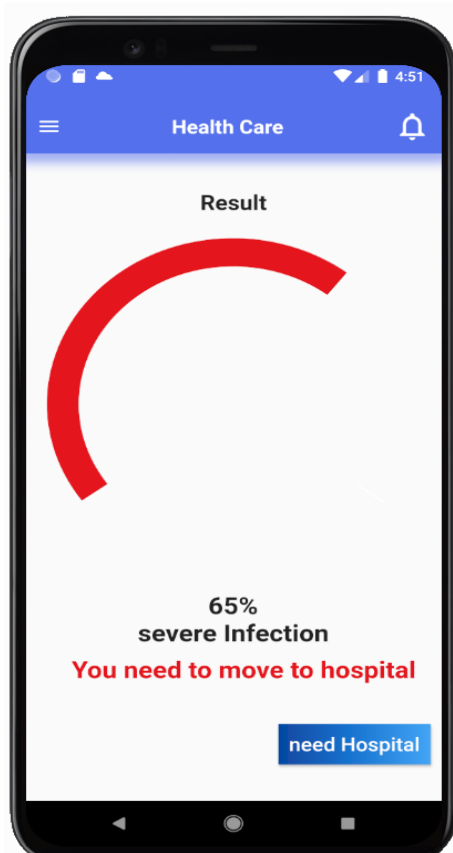


Figure 10 result screen

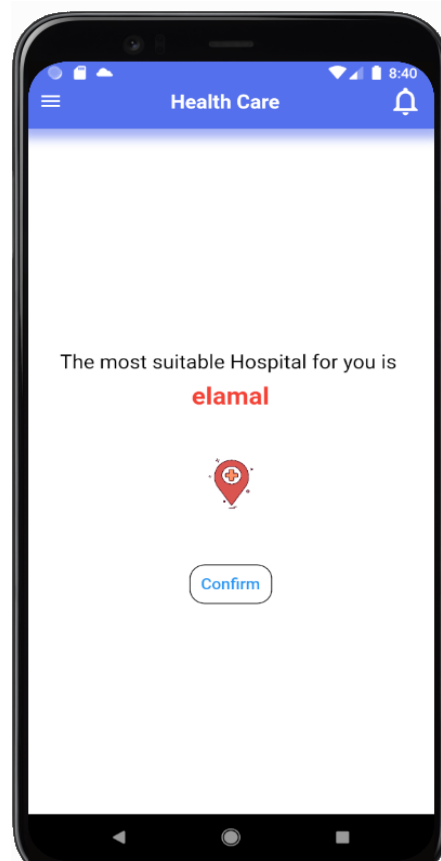


Figure 11 hospital screen

3.1.3.6 Confirm

Patient should confirm going with his national id to upload his data to database to appear in hospital application then hospital will send ambulance to take patient from home.

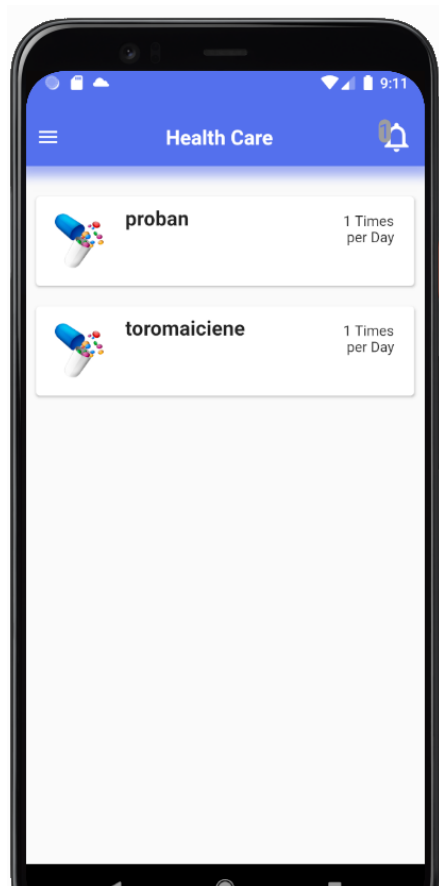


Figure 13medicaments screen

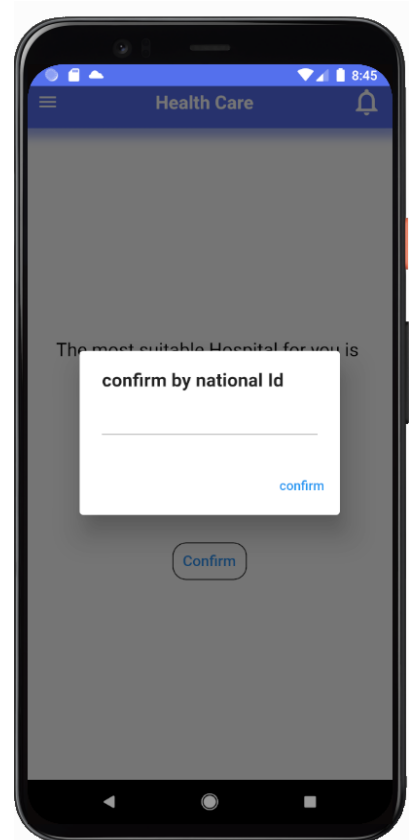


Figure 12confirm by national id

3.1.3.7 Treatment protocol

- This screen will be available during home quarantine after doctor sees patient case and add medicaments.
- Each medicament is displayed with its potion, how many times per day and any notes are written by doctor.

3.1.3.8 Chat

- If doctor see that he want to communicate with patient to know more information about his case during home quarantine he will open chat from his application (doctor application) then it will be opened in patient mobile application and they will be able to communicate with each other through this chat.
- Doctor can easily close this chat at any time he sees that he don't need it.

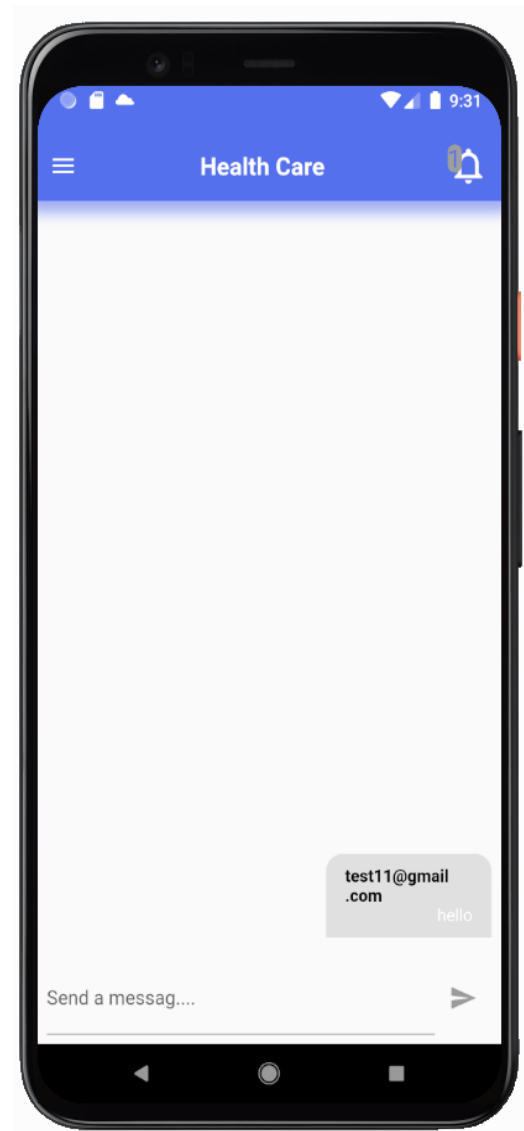


Figure 14patient chat screen

3.1.3.9 Awareness

This page displays important information about covid-19 and helps patients at quarantine by give them periodic instruction about virus.

3.1.3.10 Notification

User will receive notification if there is any change in the application like:

- Chat opened
- New message coming
- Medicaments added or modified

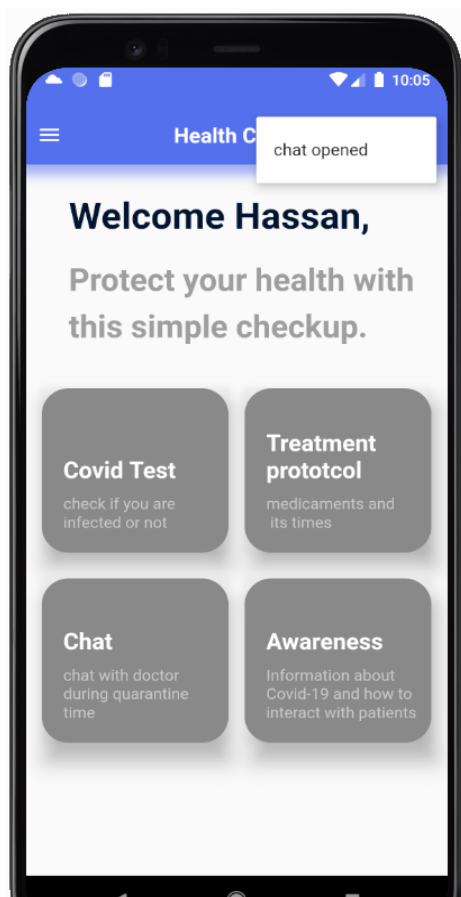


Figure 16notification

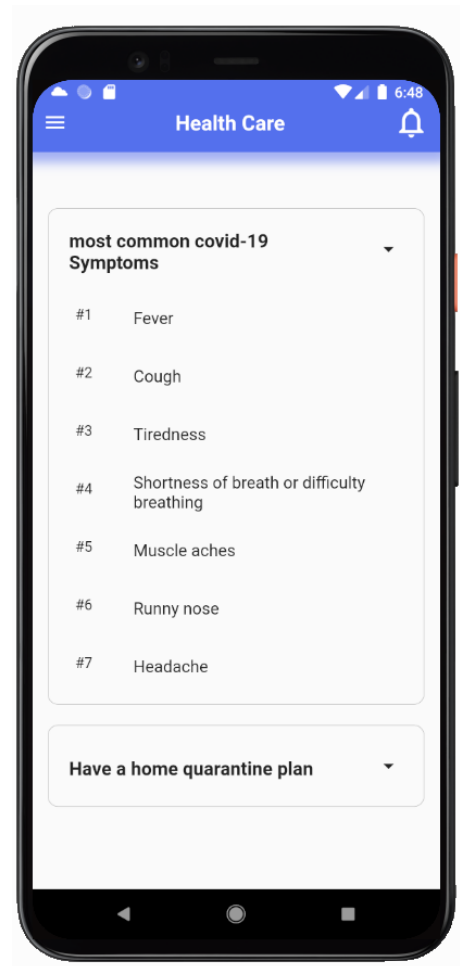


Figure 15awareness

3.1.3.11 Connect to IOT devices

During home quarantine patient will receive IOT devices keep him in touch with doctor to see if there is any dangerous change in his case so to connect this devices to the system for this patient we need to scan qr-code for the device and this will be done with patient application.

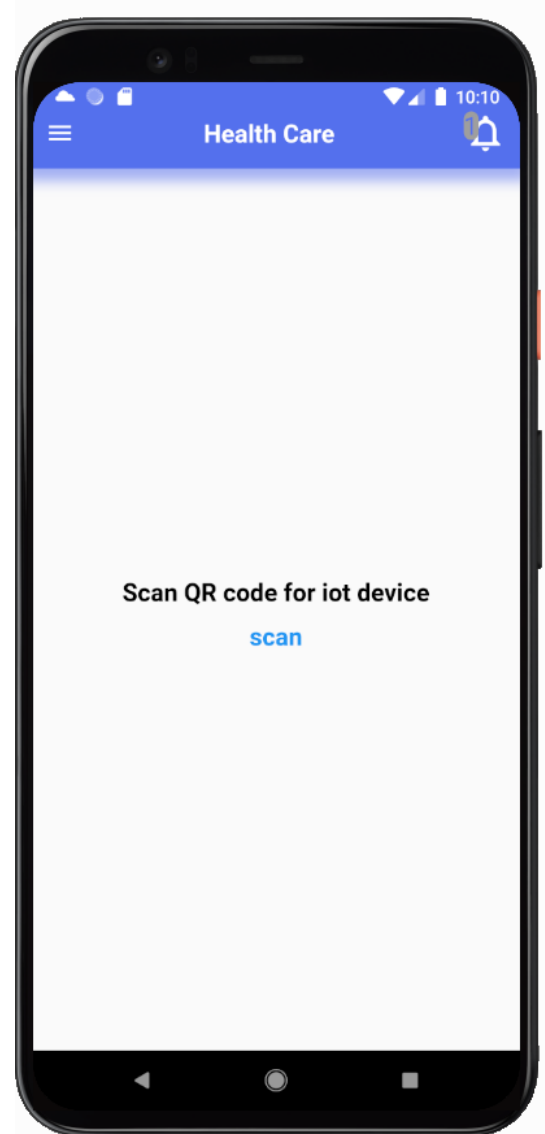


Figure 17connect iot

3.2 Doctor's mobile application

3.2.1 Flow Chart of the system

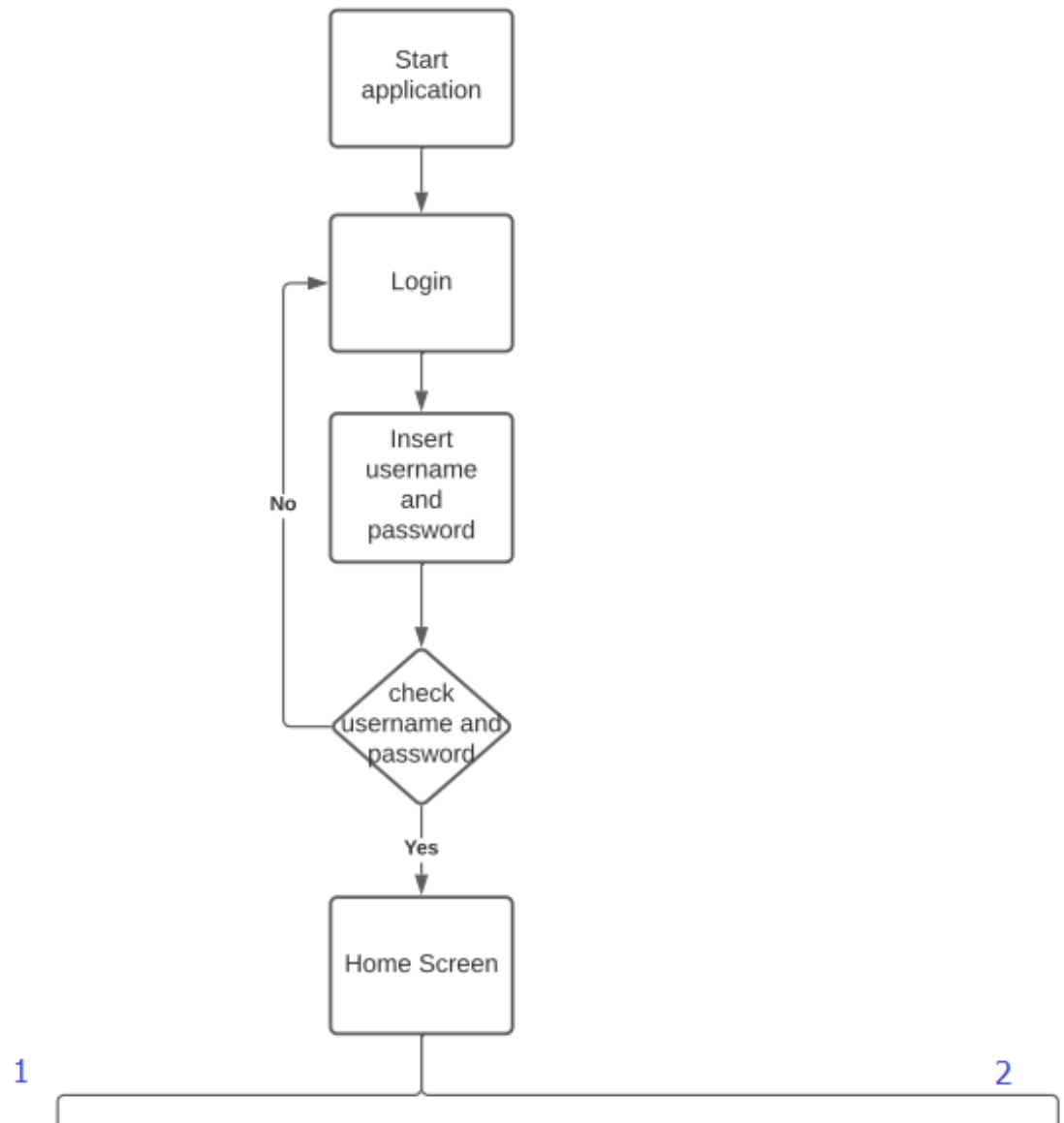
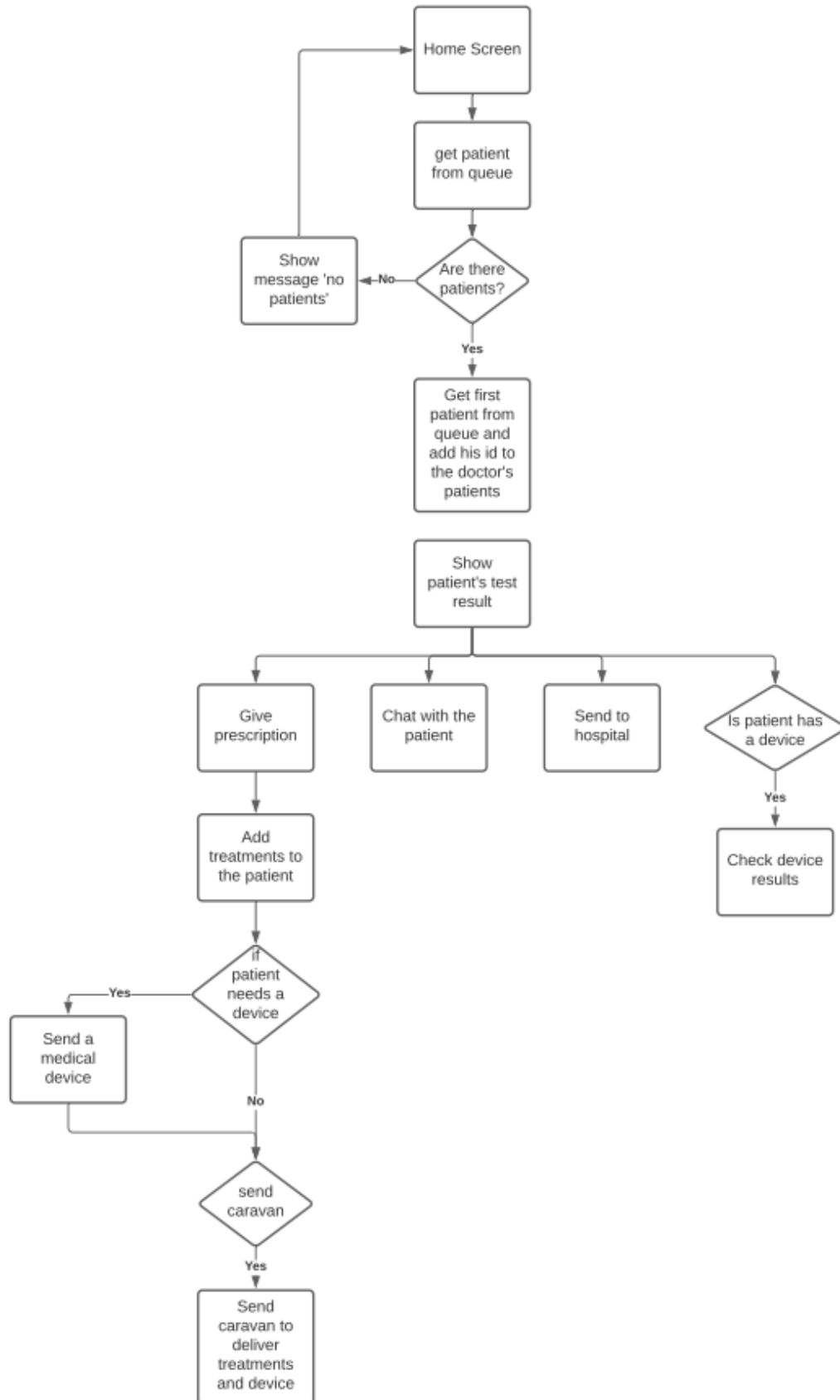
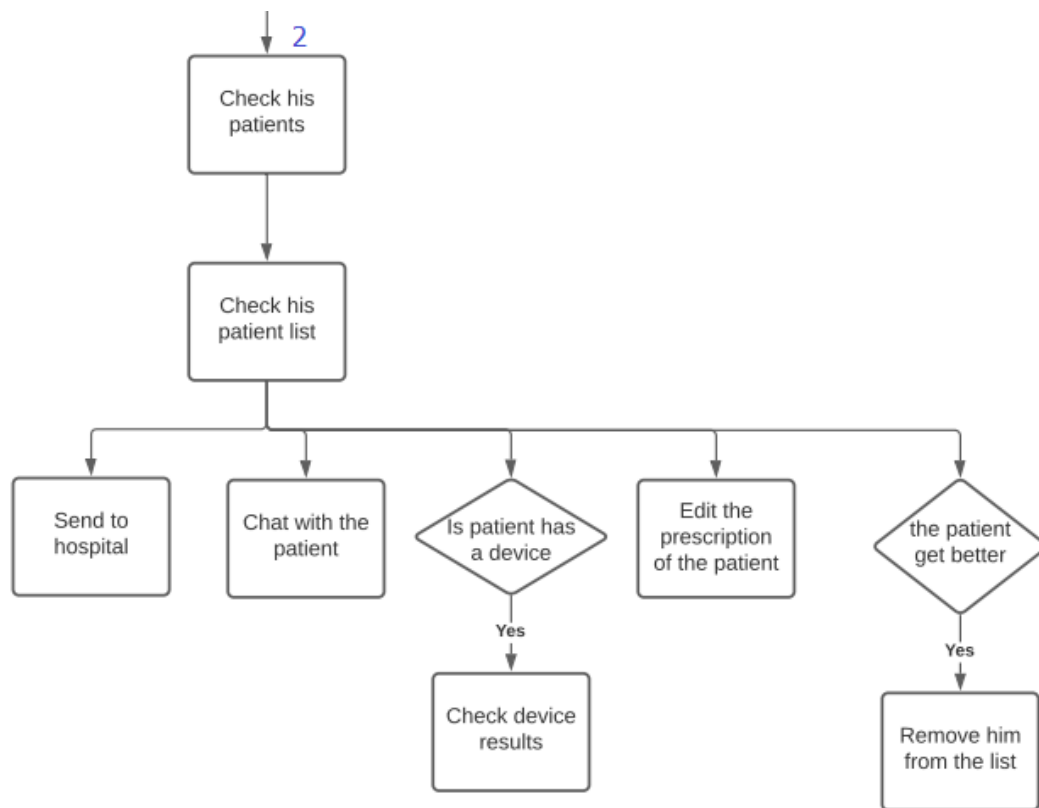


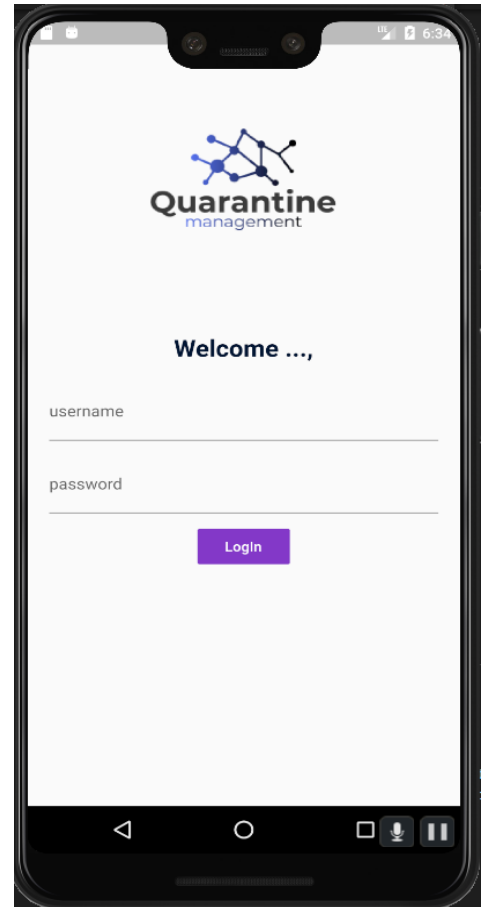
Figure 18doctor application flow chart





3.2.2 Application description

3.2.2.1 Login screen



- The user enters the username and password.
- If the username or password are incorrect, an error message will appear.
- If correct, the home page will appear.

3.2.2.2 Home Screen

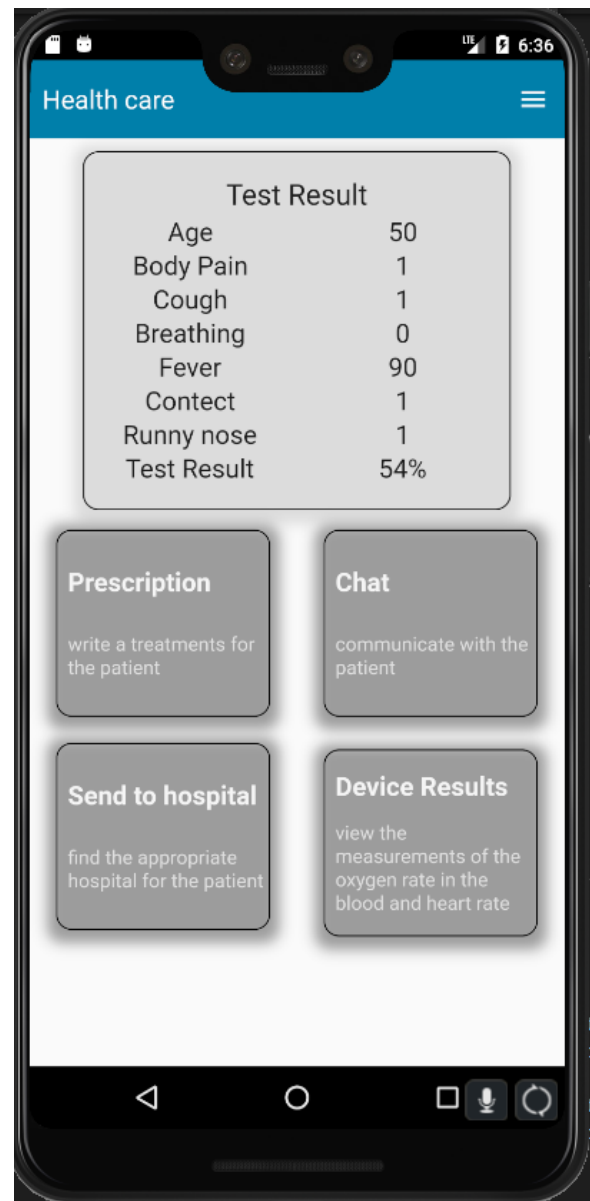


We have two options:

- a. Get next patient from the patient waiting list.
 - Get array of the patient list.
 - Take first patient and add his id to doctor's patients list to keep following up.
 - Go to patient data screen.
 - If the waiting list is empty, a message will appear to the user telling him that the list is empty.

- b. Go to user's patients screen to keep following them or unfollow when the patient gets better.

3.2.2.3 Patient data screen



The user has many options:

- a. Show the patient's test results.
- b. Give Prescribe medicine to the patient.
- c. Communicate with the patient using chat.
- d. Send to suitable hospital.

- e. If the patient has the medical device, doctor will be able to show the device result.

3.2.2.4 Prescription screen

The screenshot shows a mobile application interface for a prescription screen. At the top, there is a header bar with a status bar showing the time as 7:01. Below the header, there is a form with a list of medicines. The first medicine is 'arixtra' with a status of 'Added' and a green checkmark. Below it, there is a field for 'Day' with a dropdown arrow and a 'Note' field. The second medicine is 'actemra' with a status of 'Added' and a green checkmark. Below it, there is a field for 'Day' with a dropdown arrow and a 'Note' field. The third medicine is 'Azithromycin' with a status of 'Added' and a green checkmark. Below the list, there is a 'Send Device ?' toggle switch and a 'Submit' button. The screen is framed by a black border, and the bottom navigation bar is visible.

A list of medicines will appear to the doctor to choose which will give to the patient according to its status.

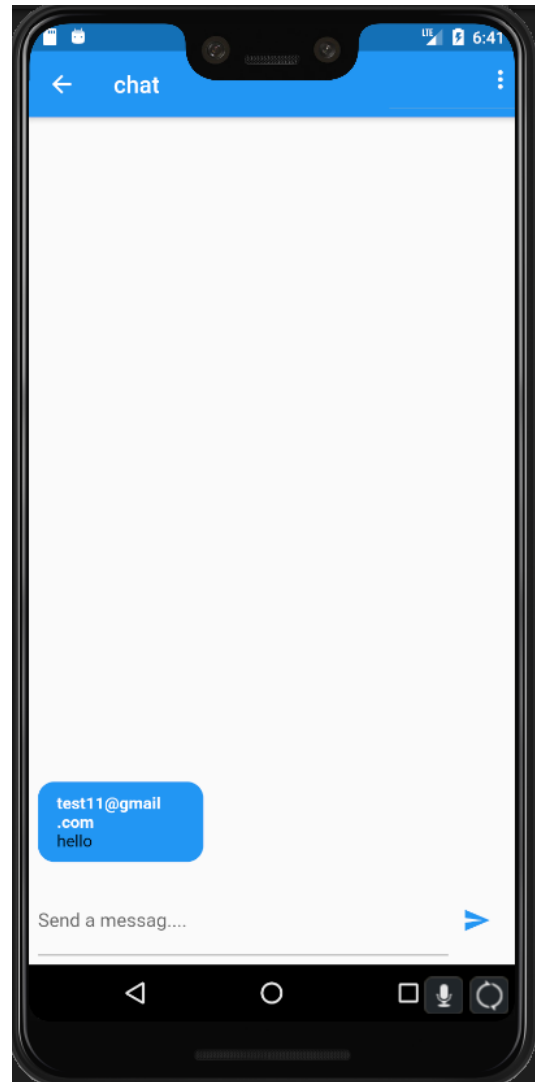
So the doctor will be able to

- a. Add medicines to the patient.
- b. Add to every medicine note and dose.
- c. Send heart beat and oxygen rate in blood device to the patient.
- d. Send caravan to deliver medicines and device to the patient.

The prescription sent to
patient,Send Caravans ?

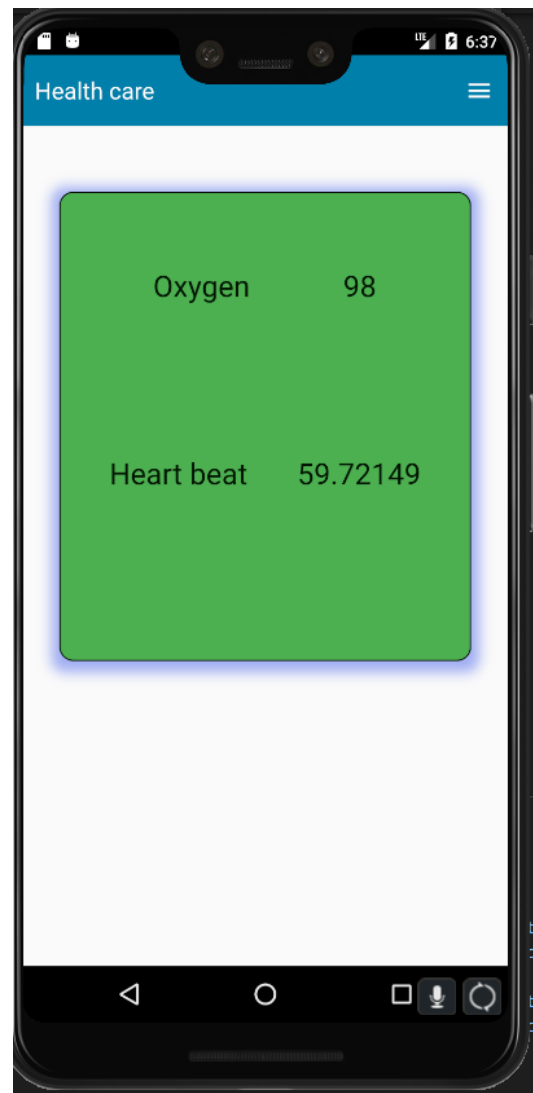
no yes

3.2.2.5 Chat screen



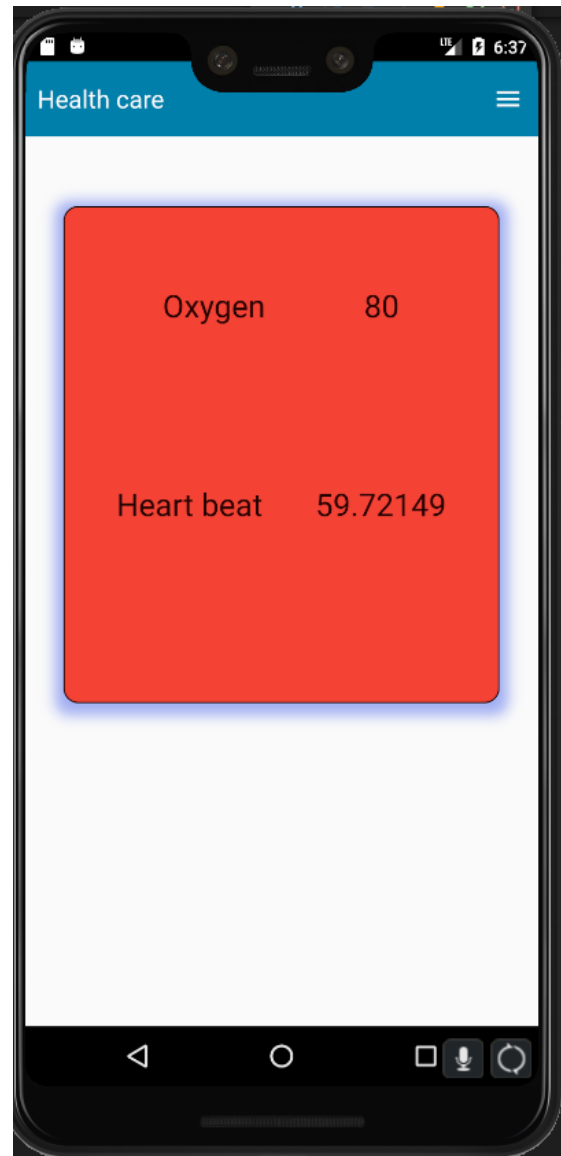
Doctor and patient can communicate with each other.

3.2.2.6 Device result screen



Doctor can show the Latest readings of heart beat and oxygen of the patient to take a decision.

- edit the Prescription.
- chat with the patient to give advices.
- send to hospital.

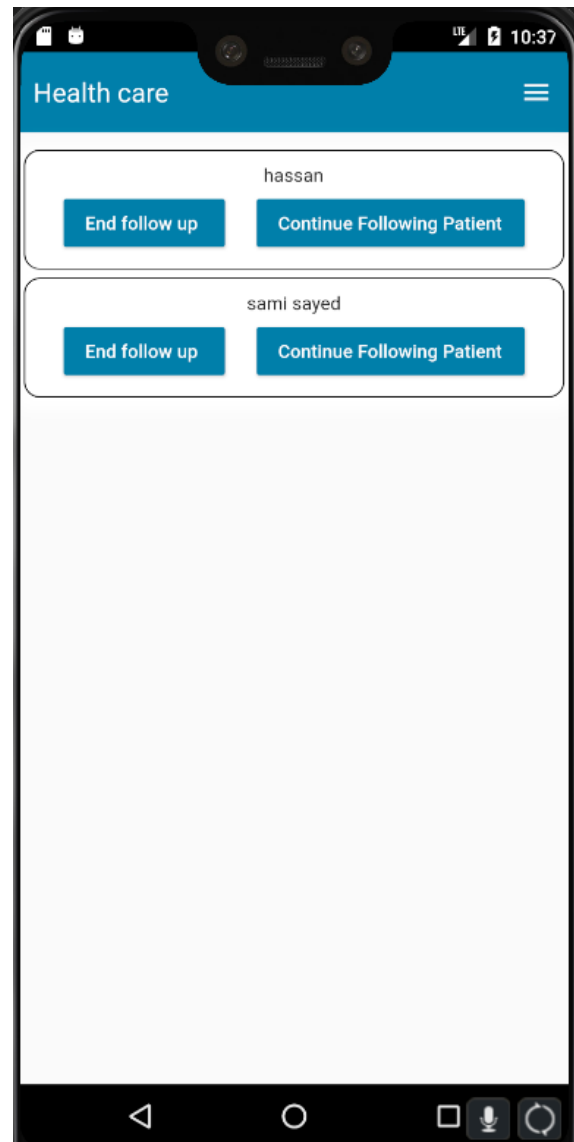


The color changes to red if the status is dangerous.

The status is normal when:

- Heart beat is in range of 40 to 60.
- Oxygen is in range of 90 to 100.

3.2.2.7 My patients screen



In this screen, doctor can see all his patients and can:

- a. End follow up a patient when he gets better by remove him from his patients list.

- b. Continue following the patient and can:
- Chat with him
 - Edit medicines.
 - Send to hospital.
 - Show oxygen rate and heart beat results.

3.3 Medical caravan application

3.3.1 Flow Chart of the system



Figure 19medical caravans flow chart

3.3.2 Application description

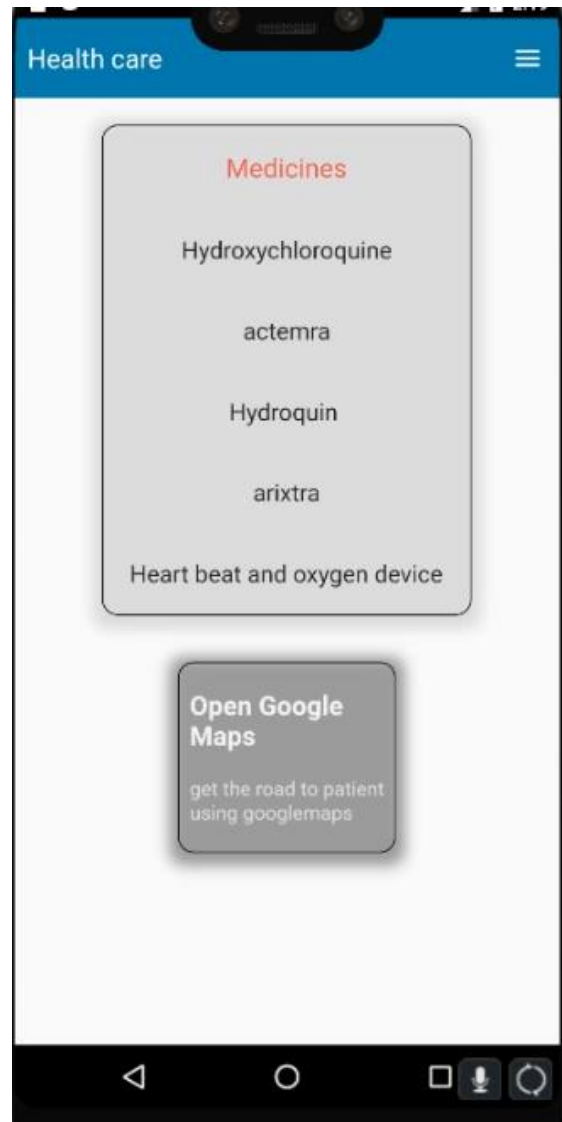
3.3.2.1 Home screen



Get next patient order to deliver the medicines by

- a. Get first patient id from the queue.
- b. Remove the id from the queue.
- c. Get patient's location.
- d. Get patient's medicines and devices.

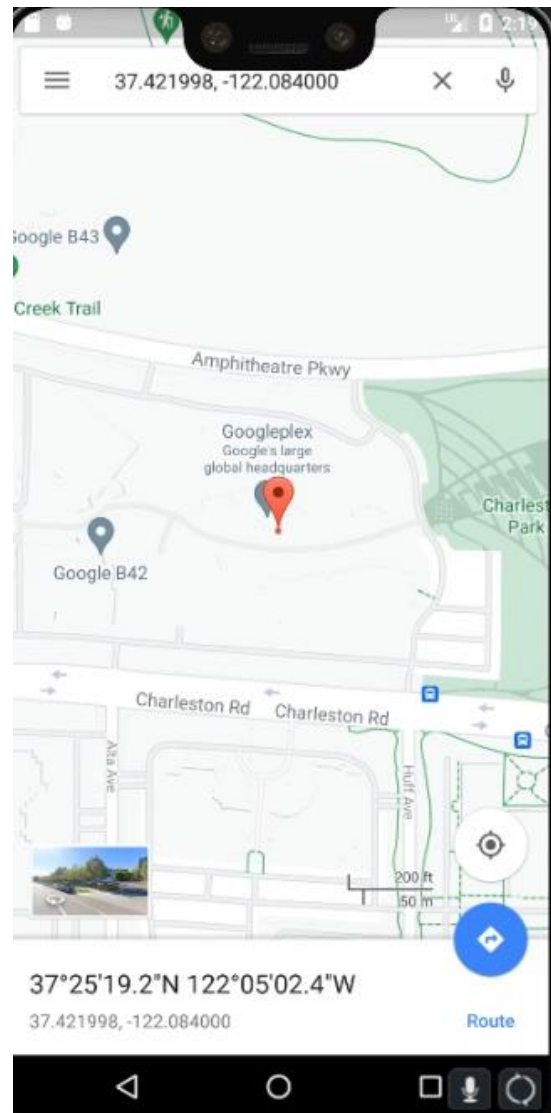
3.3.2.2 Patient data screen



User can

- Show patient's medicines.
- Show if he needs to heart beat and oxygen device or not.
- Show patient's location and open it using google maps.

3.3.2.3 Google map screen



User can get directions to deliver the order.

1.1.1 Flutter design

Flutter is Google's UI toolkit for building natively compiled applications for mobile, web, and desktop from a single codebase.

It's a cross-platform tool intended for creating Android and iOS apps from a single code base by using a modern, reactive framework. Flutter apps are built using Dart, a simple object-oriented programming language

Flutter uses Dart programming language which was introduced by Google in 2011 and is rarely used by developers. Dart syntax is easy to understand for JavaScript or Java developers as it supports most of the object-oriented concepts.

1.1.2 Benefits of Flutter:

- 1-High productivity.
- 2-Great performance.
- 3-Fast and simple development.
- 4-Compatibility.
- 5-Open-source.

1.1.3 Flutter Widgets

Everything in Flutter is a widget. Whenever you are going to code for building anything in Flutter, it will be inside a widget. The central purpose is to build the app out of widgets. It describes how your app view should look like with their current configuration and state. When you made any alteration in the code, the widget rebuilds its description by calculating the difference of previous and current widget to determine the minimal changes for rendering in UI of the app.

The image is a simple visual representation of the widget tree

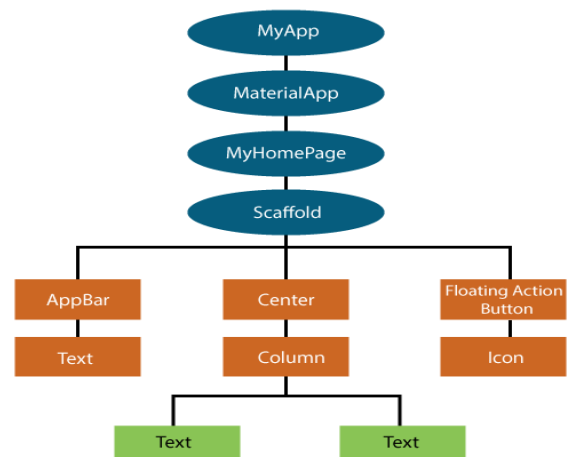


Figure 0-19(flutter widgets)

1.1.4 Flutter Layouts

The main concept of the layout mechanism is the widget. We know that flutter assume everything as a widget. So the image, icon, text, and even the layout of your app are all widgets. Here, some of the things you do not see on your app UI, such as rows, columns, and grids that arrange, constrain, and align the visible widgets are also the widgets.

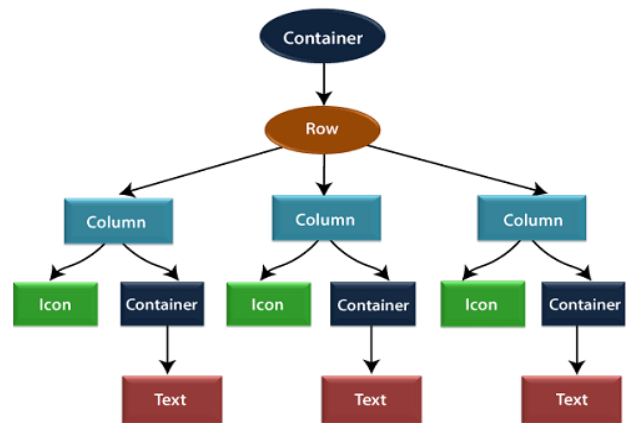


Figure 0-20 (flutter layouts)

2.1.5 Firebase:

Firebase is a platform developed by google for creating mobile and web applications. It was originally an independent company founded in 2011. In 2014 google acquired the platform and it is now their flagship offering for app development

Firebase contains 2 types of databases: Cloud firestore and Realtime database.

In our project we are using cloud firestore as database for our system

Data Structure in Firebase Firestore

How to structure data for Chat Apps in Firebase Firestore Database. Firestore is NoSQL data structure so we need to manage data relation logically and in some cases we keep some data duplication and redundancy to reduce database usage quota.

In relational database like MySQL we always taken care of Data Normalization Rules but in Firestore data structure we take care of Usage Quota which includes Number of Reads, Number of Writes and Number of Deletes.

Firestore Database consists of 3 things

1. Collections - Collections are nothing but a set of document.
2. Documents - Documents are set of sub-collections and field.
3. Fields - Field consists of attribute and value. The values are of the type of string, number, boolean, map, array, null, timestamp, geopoints and reference.

Authentication in Flutter Using Firebase

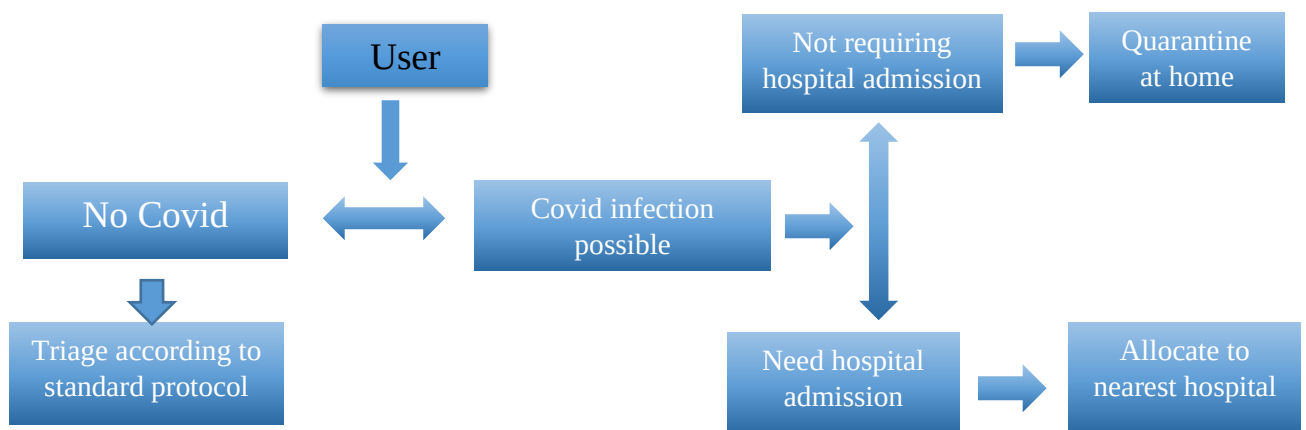
Google Firebase provides Email authentication using email. The basic plan of Firebase includes free email. We will learn Firebase Email Authentication in Flutter in this article. We will start from Firebase and will programmatically set up the actual integration in Flutter.

1.2 Flask Framework

Flask is a micro web framework written in Python. It is classified as a microframework because it does not require particular tools or libraries. It has no database abstraction layer, form validation, or any other components where pre-existing third-party libraries provide common functions. However, Flask supports extensions that can add application features as if they were implemented in Flask itself. Extensions exist for object-relational mappers, form validation, upload handling, various open authentication technologies and several common framework related tools.

In our project we used flask for:

- Building local server to run algorithms and do different processes on data.
- Create rest-api to connect mobile application with backend services.
- We use http requests (get ,post ,update ,delete).



3.4 codes

```
class CovidDetect(Resource):
    def post(self):
        x = 0
        args = test_args.parse_args()
        if args['diffbreathing'] == 1 and args['cough'] == 1 and args['cough'] == 1:
            x = x+10
        inputfeatures = [args["fever"], args["age"],
                        args["bodypain"], args["cough"], args["diffbreathing"], args['Gender'], args['cough'], args['Runnynose'], args['
        infprob = clf.predict_proba(inputfeatures)[0][1]
```

Figure 22 flask codes

```
import pickle
from flask import Flask
from flask_restful import Api, Resource, reqparse, abort
from temp import Dict
from firebase_admin import credentials, firestore
from firebase_admin import db
import firebase_admin
import math

app = Flask(__name__)
api = Api(app)

file = open('model_pickle', 'rb')
clf = pickle.load(file)

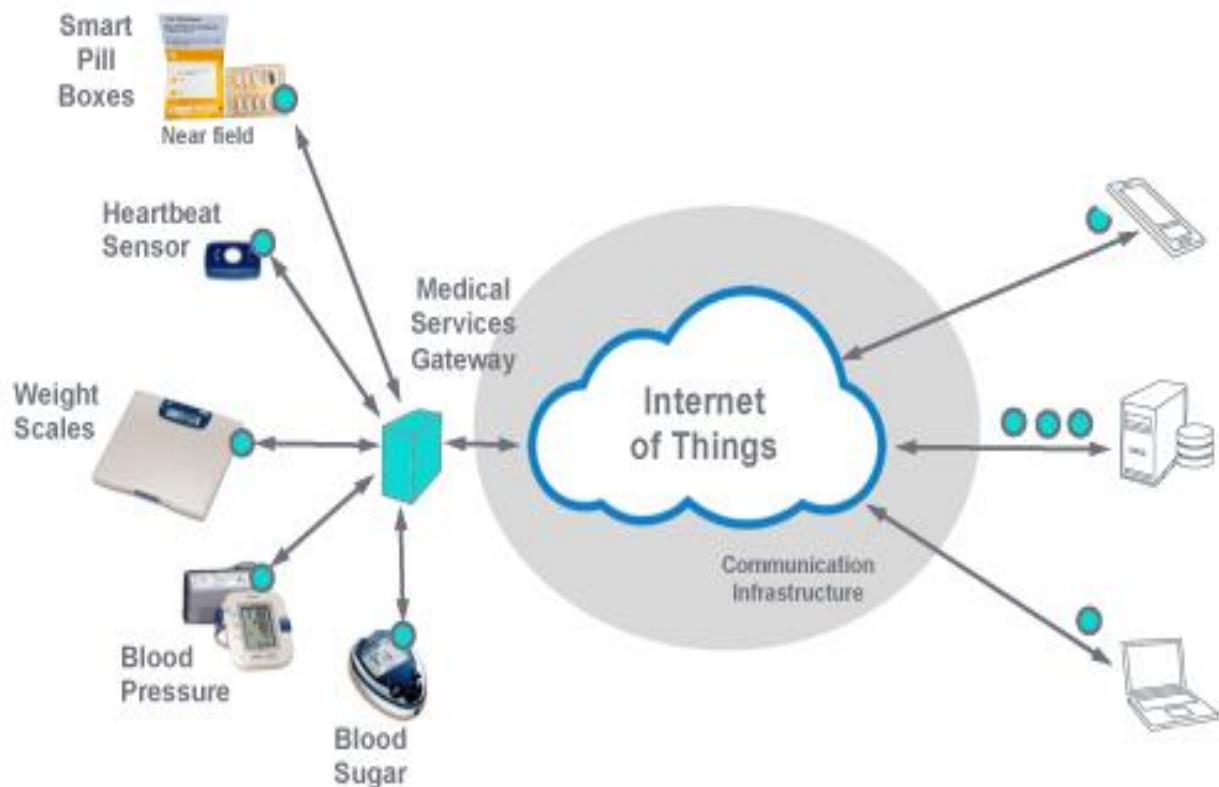
res_args = {}
test_args = reqparse.RequestParser()
test_args.add_argument("fever",
                        required=False, type=int, help='fever degree is required')
test_args.add_argument("age", required=True, type=int, help='age is required')
test_args.add_argument(
    "bodypain", help="bodypain is required", required=True, type=int)
test_args.add_argument("cough", help="cough is required",
                        required=False, type=int)
test_args.add_argument(
    "diffbreathing", help="diffbreathing is required", required=True, type=int)
```

```

122 class Allocation(Resource):
123     allocation_args = reqparse.RequestParser()
124     allocation_args.add_argument("lat",
125                                 required=False, type=float, help='lat is required')
126     allocation_args.add_argument("long",
127                                 required=False, type=float, help='long is required')
128     allocation_args.add_argument("cond",
129                                 required=False, type=int, help='cond is required')
130
131     def post(self):
132         args = Allocation.allocation_args.parse_args()
133         result = Find(args['cond'], args['lat'], args['long'])
134         return result
135
136 api.add_resource(Allocation, "/allocation")
137 api.add_resource(CovidDetect, "/covid")
138 # api.add_resource(Allocation, "/allocation")
139
140 if __name__ == "__main__":
141     app.run(host='0.0.0.0', debug=True)

```

Chapter 4: IOT



4.1 Internet of Things

This project is made for doctors who want to see their corona virus patient status so the patient will use the device to measure his temperature and his heart beat per minute and spo2 percentage and the program will do the rest such as sending patient status data through the internet to the application which will check the incoming data and see if the patient has temperature more than 38 and his heart beat average is more than 100 or less than 60 and his spo2 is less than 90% then patient status is critical. So doctor who is responsible for this patient will get a warning notification letting him know that this patient need an ambulance.

When the world suffer from covid-19 someone has to figure some solutions for helping people so we made our researches and see what can we do for that virus. We used the new technology (IoT) for letting doctors communicate with their patients and get the data which they need like body temperature, heart beat per minute, and spo2 percentage. By using the internet of things technology letting the sensors and other devices connect to the internet through Wi-Fi

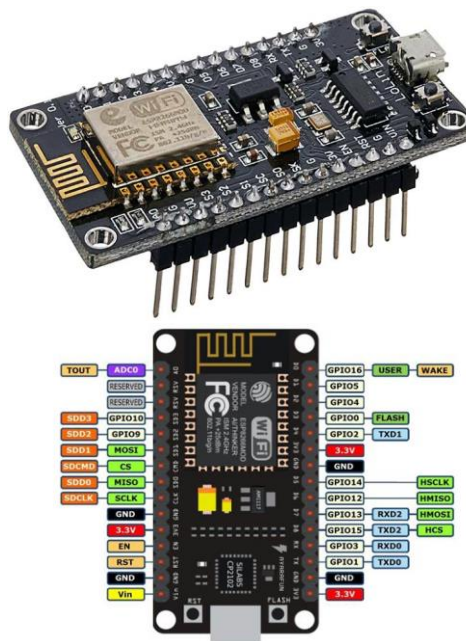
4.2 Project Specification

Components:

Node Mcu esp8266 :

Is a development board specially targeted for IoT based Applications, It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems.

This is the control unit which can be programmed for a specific purpose and we will use it to get the measurement values and send it to the application through the internet



it has a Wi-Fi chip built in which make it easier to use Wi-Fi and connect the sensors to the internet and controlled through the internet too.

This chip programmed by Arduino IDE and with Arduino c language.

-Nodemcu esp8266 specifications:

- Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106

- Operating Voltage: 3.3V

- Input Voltage: 7-12V

- Digital I/O Pins (DIO): 16

- Analog Input Pins (ADC): 1

- UARTs: 1

- SPIs: 1

- I2Cs: 1

- **Mlx90614 Infrared sensor:**

it is a temperature measurement device works on infrared radiations. It is the best option for such applications where we cannot easily reach to measure temperature because it provides non-contact measurement of temperature.

- It has an amplifier of squat noise, Analog to Digital (ADC) converter of 17-bits and potent DSP component, all these components provides the elevated exactness and high degree of this thermometer.

- During its fabrication, it is mounted with the 10 bit Pulse width modulation (PWM) which constantly sends out the calculated temperature in the assortment of -20 to 120 °C by providing yield resolution of 0.14 °C.

- It consists of two boards (chips) one is infrared radiation detector and other is ASSP which is a signal conditioner, it is for the processing of infrared sensor output.

- The pinout of Pulse width Modulation can also be used as a thermal type of relay, which provides us less expensive completion in temperature observant applications such as boiling and freezing.

- This device has two input supply choices one is 5V and other is the 3V battery. We can supply 5 volts by exterior supply.



- **MLX90614 Specifications:**
- Operating Voltage: 3.6V to 5V (available in 3V and 5V version)
- Supply Current: 1.5mA
- Object Temperature Range: -700 C to 382.20C
- Ambient Temperature Range: -400 C to 1250C
- Accuracy: 0.020C
- Field of View: 800
- Distance between object and sensor: 2cm-5cm (approx.)

Max30100 Sensor:

The MAX30100 is an integrated pulse oximetry and heartrate monitor sensor solution.

It combines two LEDs, a photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals.

It breakout operates from 1.8V and 5.5V.

-The device has two LEDs, one emitting a red light, another emitting infrared light. For pulse rate, only the infrared light is needed.

Both the red light and infrared light is used to measure oxygen levels in the blood.

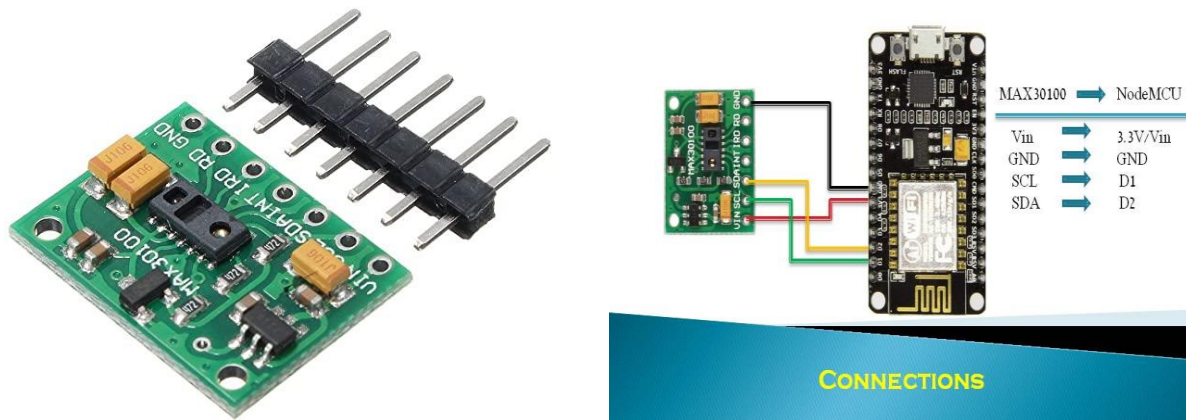
When the heart pumps blood, there is an increase in oxygenated blood as a result of having more blood.

As the heart relaxes, the volume of oxygenated blood also decreases. By knowing the time between the increase and decrease of oxygenated blood, the pulse rate is determined.

It turns out, oxygenated blood absorbs more infrared light and passes more red light while deoxygenated blood absorbs red light and passes more infrared light.

4.3 This is the main function of the MAX30100: it reads the absorption levels for both light sources and stored them in a buffer that can be read via I2C.

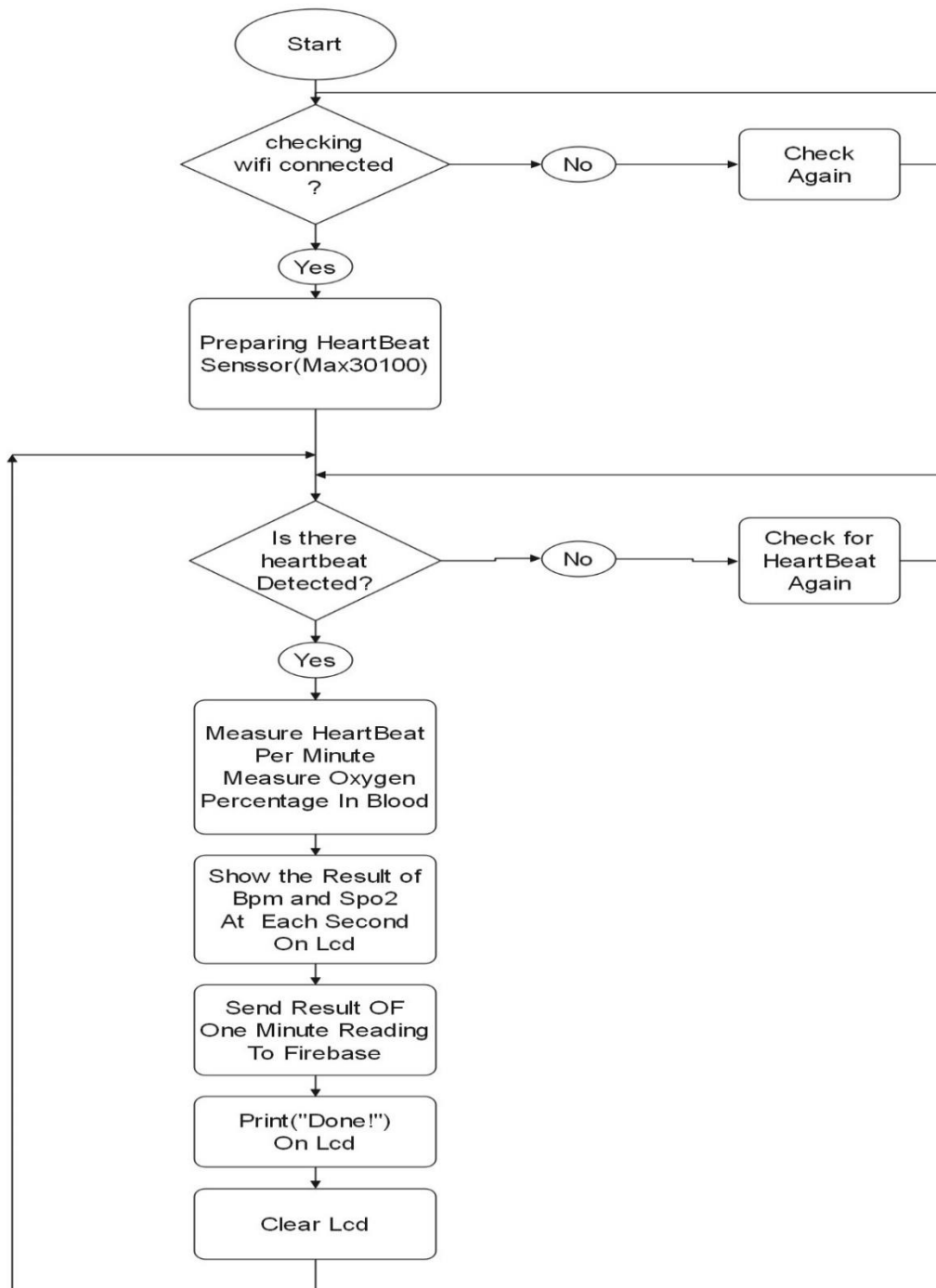
sasdd



Max30100 Specifications:

Working voltage	1.8 ~ 5.5V
Shutdown Current	0.7$\hat{A}$$\mu$A
Length (mm)	19
Width (mm)	14.5
Height (mm)	3
Weight (gm)	2
Color	Green
Shipment Weight	0.005 kg
Shipment Dimensions	8 × 6 × 4 cm

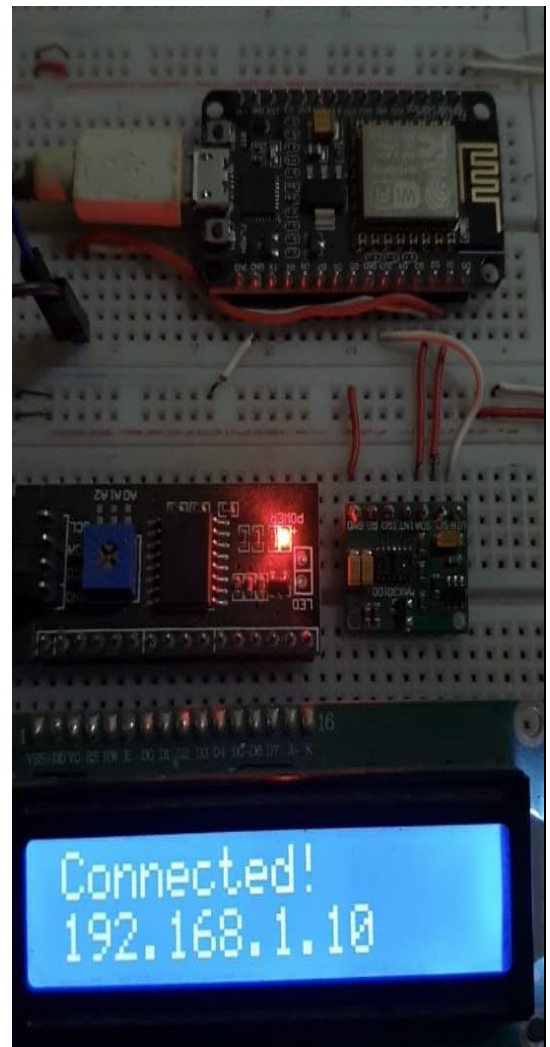
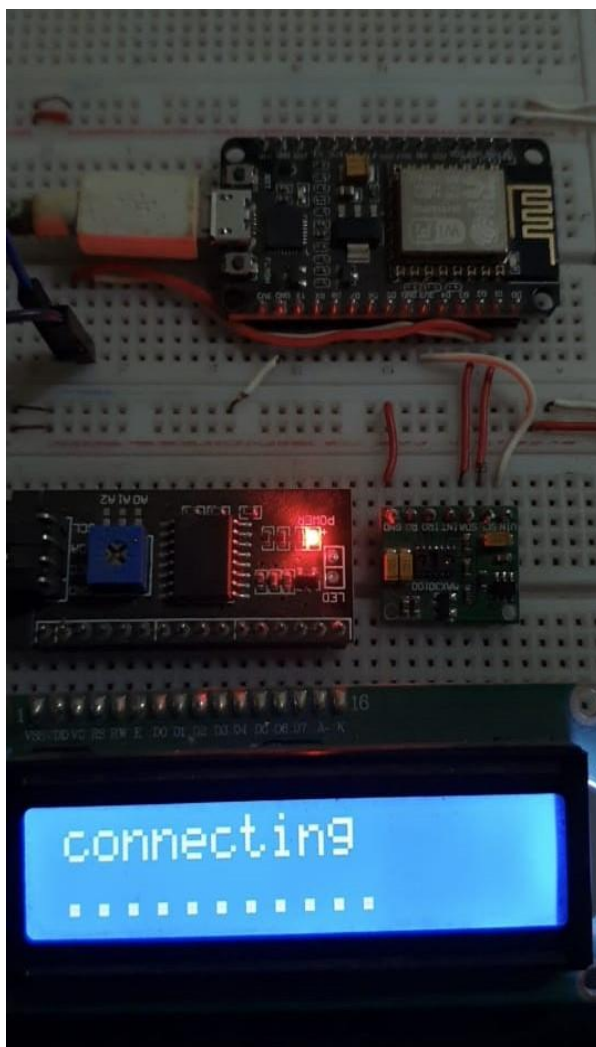
4.4 Flow Chart of IOT System



4.5 Start

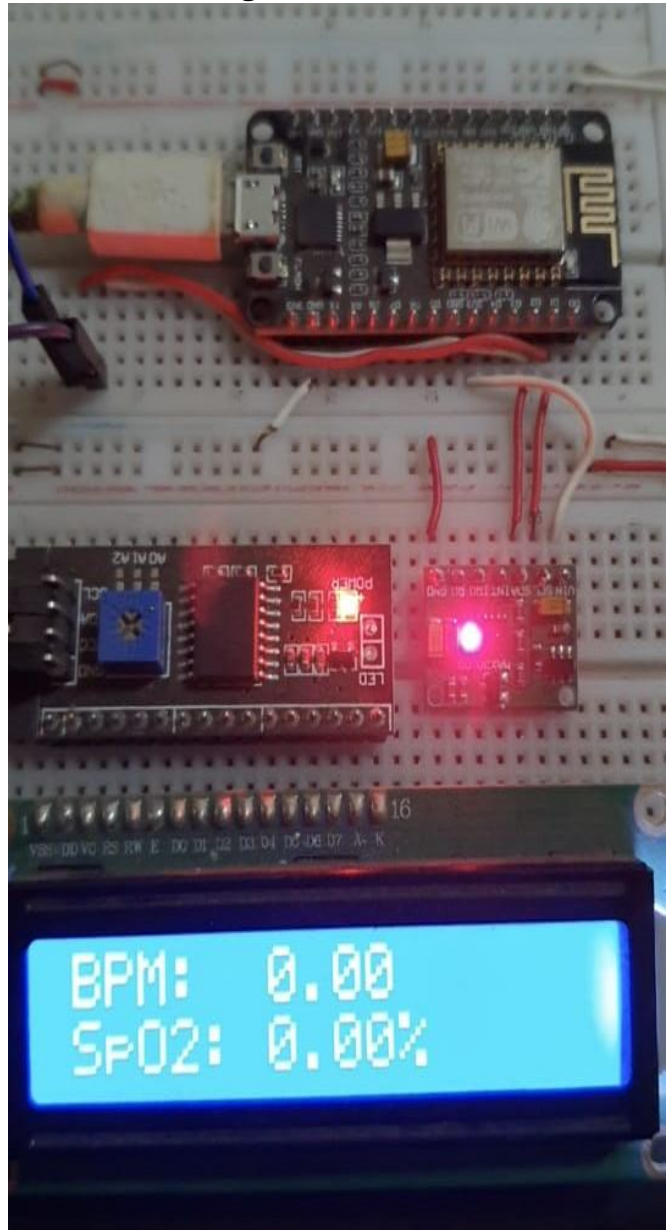
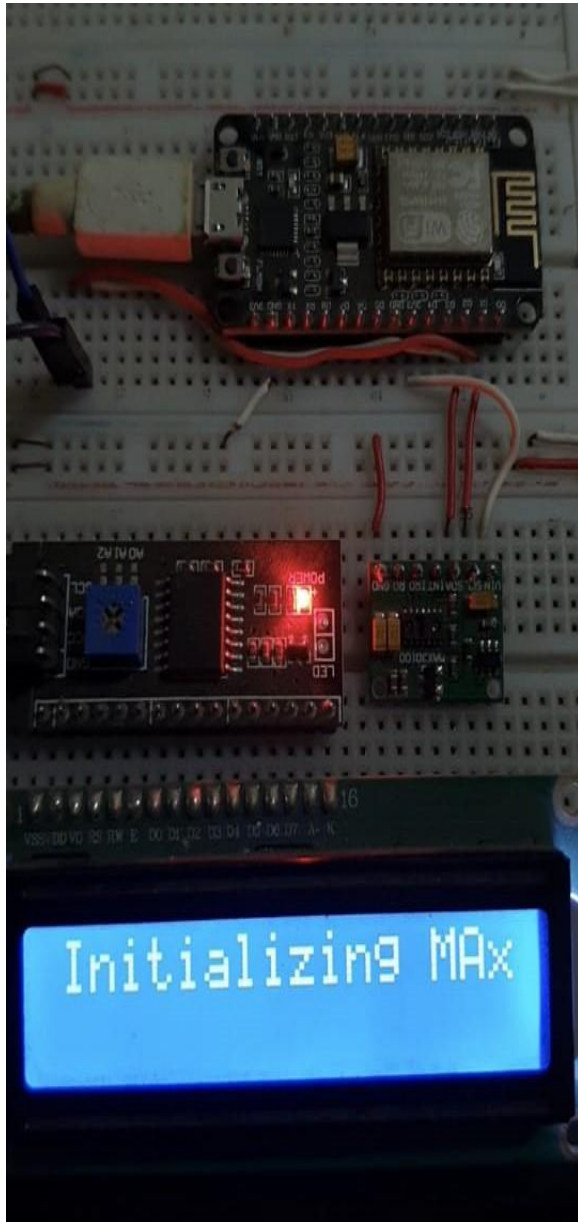
Step 1:

Program starts directly connecting Wi-Fi by the name and password written in code and prints the status on lcd then if it connected successfully prints Ip on lcd if not connected it stays in connecting page till it connect or till I repass the right name and password.



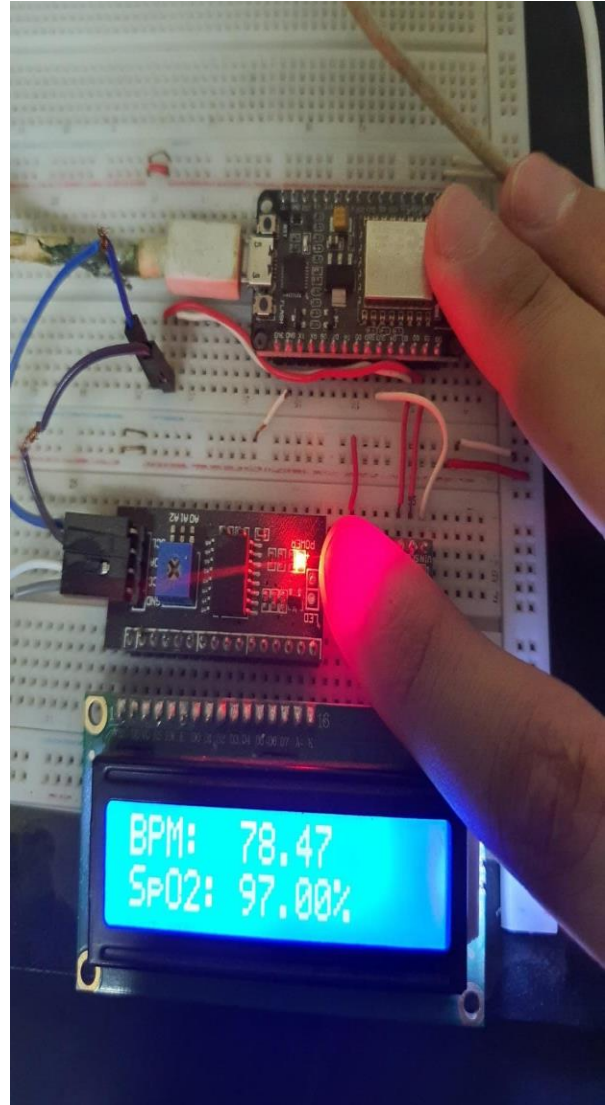
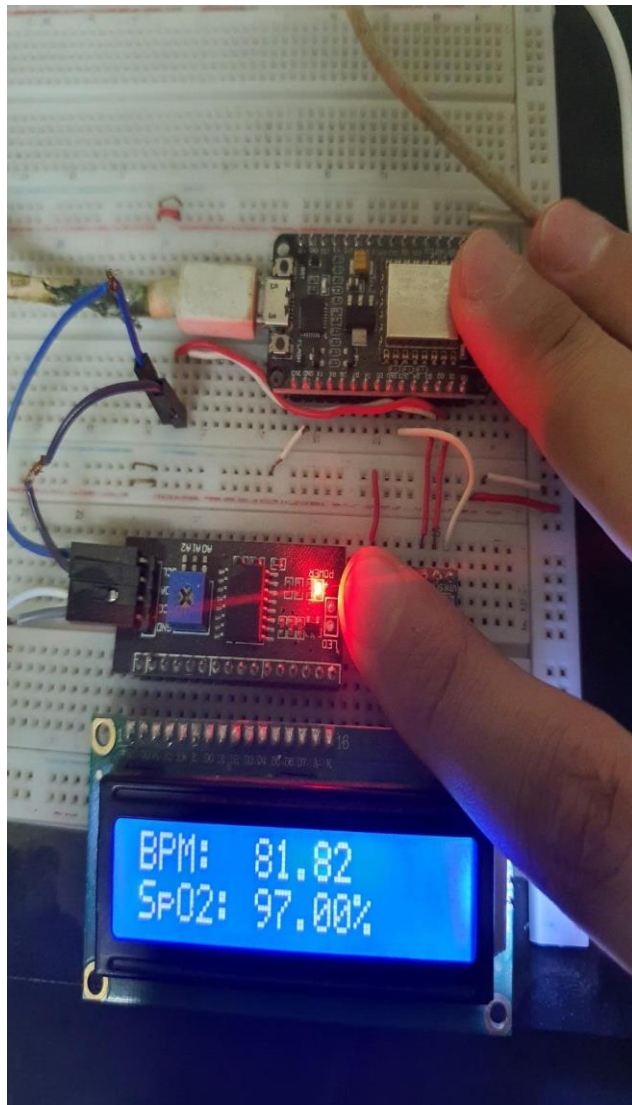
Step 2

Initializing Heartbeat sensor, after initializing sensor led illuminates as follow
during initializing after initializing



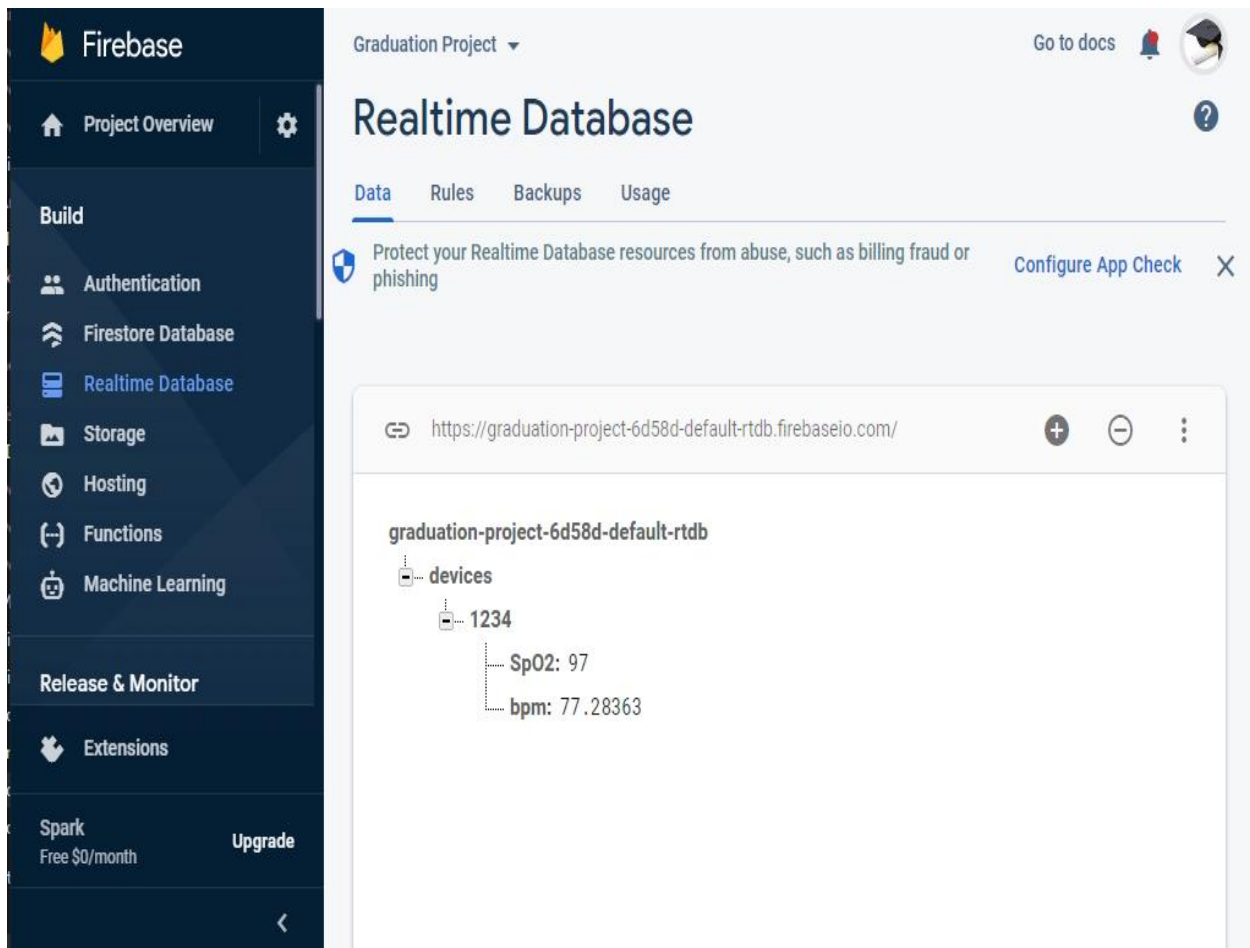
Step 3

Detecting the Heartbeat, if there is a Heartbeat Detected then
Start Measuring Heartbeat per minute and Oxygen percentage in blood
And at each second print the result on the screen as follow



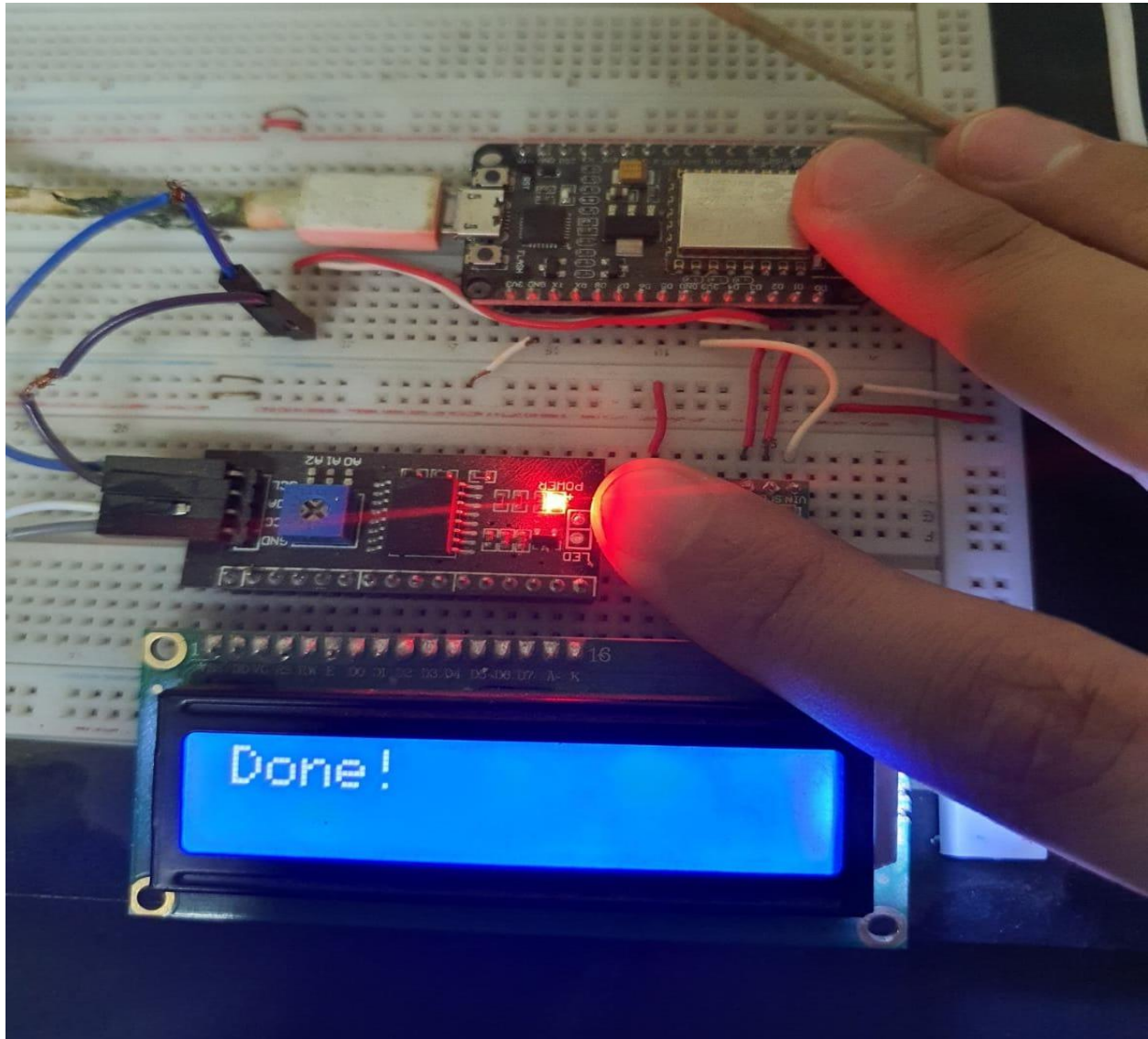
Step 4

Calculating the average of 20 reading to get the actual value of Bpm and Spo2 and send the result to Firebase cloud letting the mobile application catching the result and view it to patient's Doctor.



Step 5

After sending process ends program will print done on lcd notifying the patient that he can release his finger.



Step 6

Redo the steps from step 3 to step 6 in Looping till the controller turned off.

Chapter 5

MACHINE LEARNING & Bed Capacity Management



In this Chapter, we propose a machine-learning model that predicts a probability Covid-19 infection by asking eight basic questions. And manage the distribution and allocation of patients over hospitals bed capacity management algorithm.

5.1 Machine Learning Model

Machine learning is a subfield of computer science that evolved from the study of pattern recognition and computational learning theory in artificial intelligence. Machine learning explores the construction and study of algorithms that can learn from and make predictions on data. Such algorithms operate by building a model from example inputs in order to make data-driven predictions or decisions, rather than following strictly static program instructions.

Machine learning is a rapidly growing field that attempts to extract general concepts from large datasets, commonly in the form of an algorithm that predicts an outcome (commonly referred to as a predictive model or estimator) a task that has become increasingly difficult to accomplish by humans because data volume and complexity has increased beyond what was capable with traditional statistics and desktop computers

5.1.1 Flow Diagram to build ML model

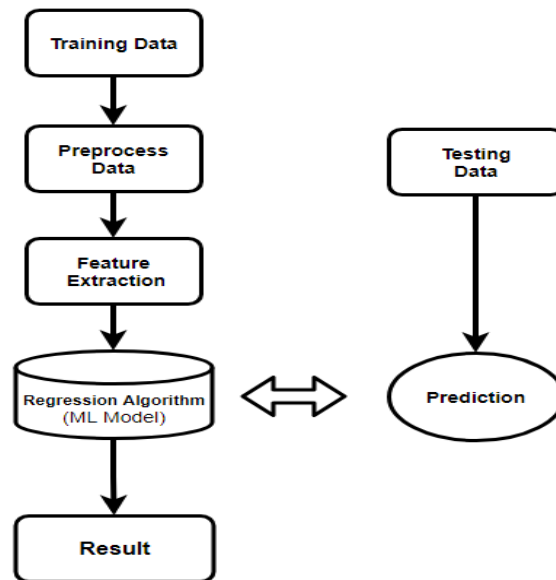


Figure 5-1 (ML flow diagram)

5.1.2 Training Data

With data are generated contain nine major variables, the description of each variable is as follow:

Fever (body temperature), Age, body pain, Cough, difficult breathing, contact (Has the person contacted some other COVID-19 Patient), Gender (Male/Female), Runny-Nose, Nasal-congestion, Covid19 (Positive/negative).

With all these categorical variables, a combination for each label in the variable will be generated and therefore, in total 2008 combinations are created

The task of the dataset is to predict probability infection coronavirus.

Table shows. The first five rows of the dataset

Out[5]:

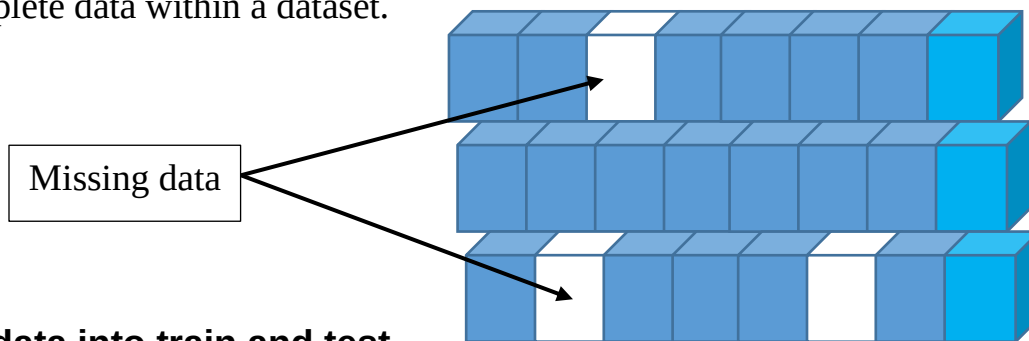
	fever	age	bodypain	cough	diffbreathing	Gender	Contact	Runny-Nose	Nasal-Congestion	Covid19
0	104	81	1	1	0	0	-1	1	0	1
1	104	2	1	1	-1	1	-1	0	1	0
2	99	69	0	1	-1	1	0	1	1	0
3	102	10	0	1	0	0	0	0	1	0
4	100	64	0	1	1	0	1	0	1	0
5	100	95	0	0	1	1	1	1	1	1
6	103	92	0	1	0	1	-1	1	0	1
7	98	36	1	0	-1	0	-1	1	0	0
8	103	25	1	1	1	0	0	0	0	1
9	104	35	0	0	-1	1	1	0	1	0

Figure 5-2(dataset table)

5.1.3 Preprocess

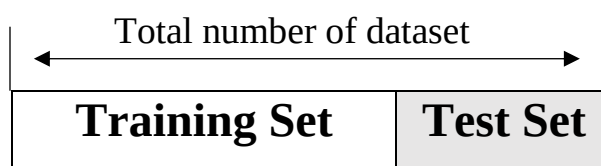
Cleaning Data

Data cleaning is one of the most important things for machine learning, it is the process of fixing or removing incorrect, corrupted, incorrectly formatted, or incomplete data within a dataset.

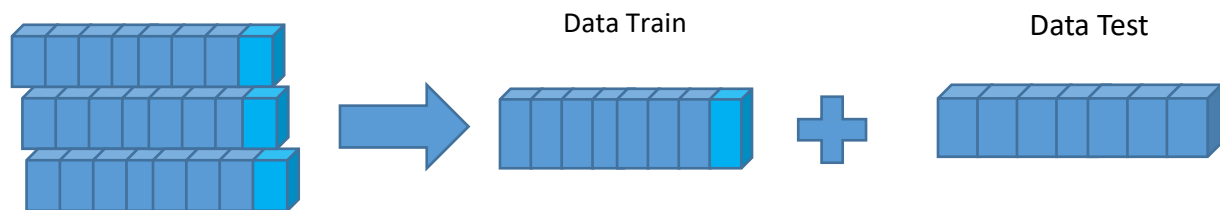


Split data into train and test

The data we use is usually split into training data and test data. The training set contains a known output and the model learns on this data in order to be generalized to other data later on. We have the test dataset in order to test our model's prediction on this subset.



To do this in python. We'll using the Scikit-Learn Library and specifically the `train_test_split` method.



5.1.4 Logistic Regression Algorithm

Logistic regression was used in the biological sciences in early twentieth century. It was then used in many social science applications. Logistic regression is used when the dependent variable (target) is categorical.

For example,

- To predict whether any person is having disease or not (1) or (0).
- To predict whether an email is spam (1) or (0).

Consider a scenario where we need to classify whether any person is having a coronavirus disease or not based on some pre-defined standard symptoms.

If we use linear regression for this problem, there is a need for setting up a threshold based on which classification can be done. Say if the actual class is positive, predicted continuous value 0.4 and the threshold value is 0.5, the data point will be classified as not positive which can lead to serious consequence in real time.

From this example, it can be inferred that linear regression is not suitable for classification problem. Linear regression is unbounded, and this brings logistic regression into picture. Their value strictly ranges from 0 to 1.

Model:

Output = 0 or 1

Hypothesis => $Z = \theta_0 + \theta_1 * X$

$h_{\theta}(X) = \text{sigmoid}(Z)$

$$\text{sigmoid}(\eta) = \frac{1}{1 + \exp(-\eta)}$$

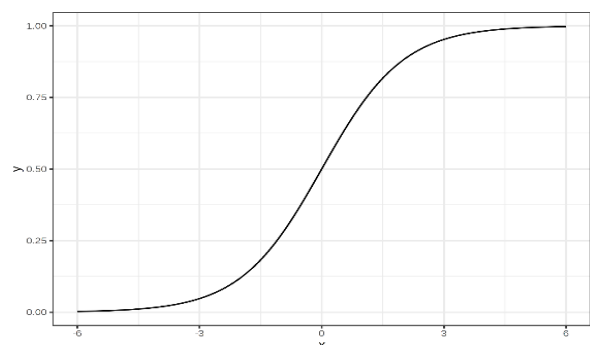
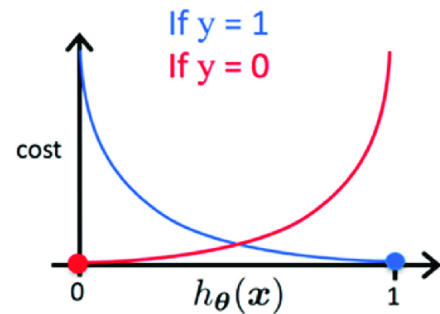


Figure 0-3 (sigmoid function)

if Z goes to infinity, Y_{pred} will become 1 and if Z goes to negative infinity, Y_{pred} will become 0.



Analysis of the hypothesis

The output from the hypothesis is the estimated probability. This is used to infer how confident can predicted value be actual when given an input X .

Cost function

The cost function represents optimization objective, and minimize it so that we can develop an accurate model with minimum error.

$$\text{cost}(h_{\theta}(x), Y(\text{actual})) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

How do we reduce the cost value? Well, this can be done by using Gradient Descent. The main goal of Gradient descent is to minimize the cost value $J(\theta)$.

Gradient descent

Now to minimize our cost function we need to run the gradient descent function on each parameter.

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

θ : parameter of x

j : number of features

α : leaning step

Start with an initial value for (θ) , and then repeat the equation with new values of (θ) . Until reach to the bottom of the hill.

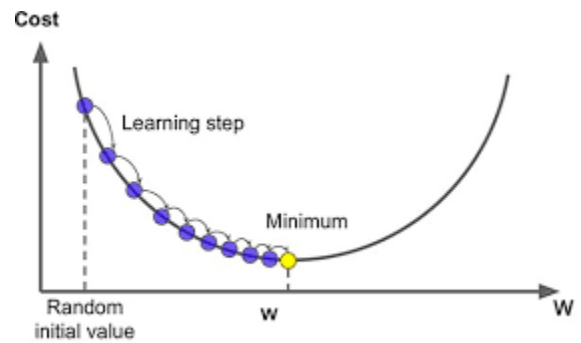


Figure 0-4 (cost function))

5.2 Bed Capacity Management

The goal of this phase is to build a decision support model that is powerful and effective. When the patient is need to quarantine at hospital the application selects all hospitals in his city, and then using sequential search algorithm to select the hospitals that are available based on free intensive care unit, and free beds. After selecting list with available hospital then using shortest path first algorithm to select the nearest hospital from his location.

The patient can view the following:

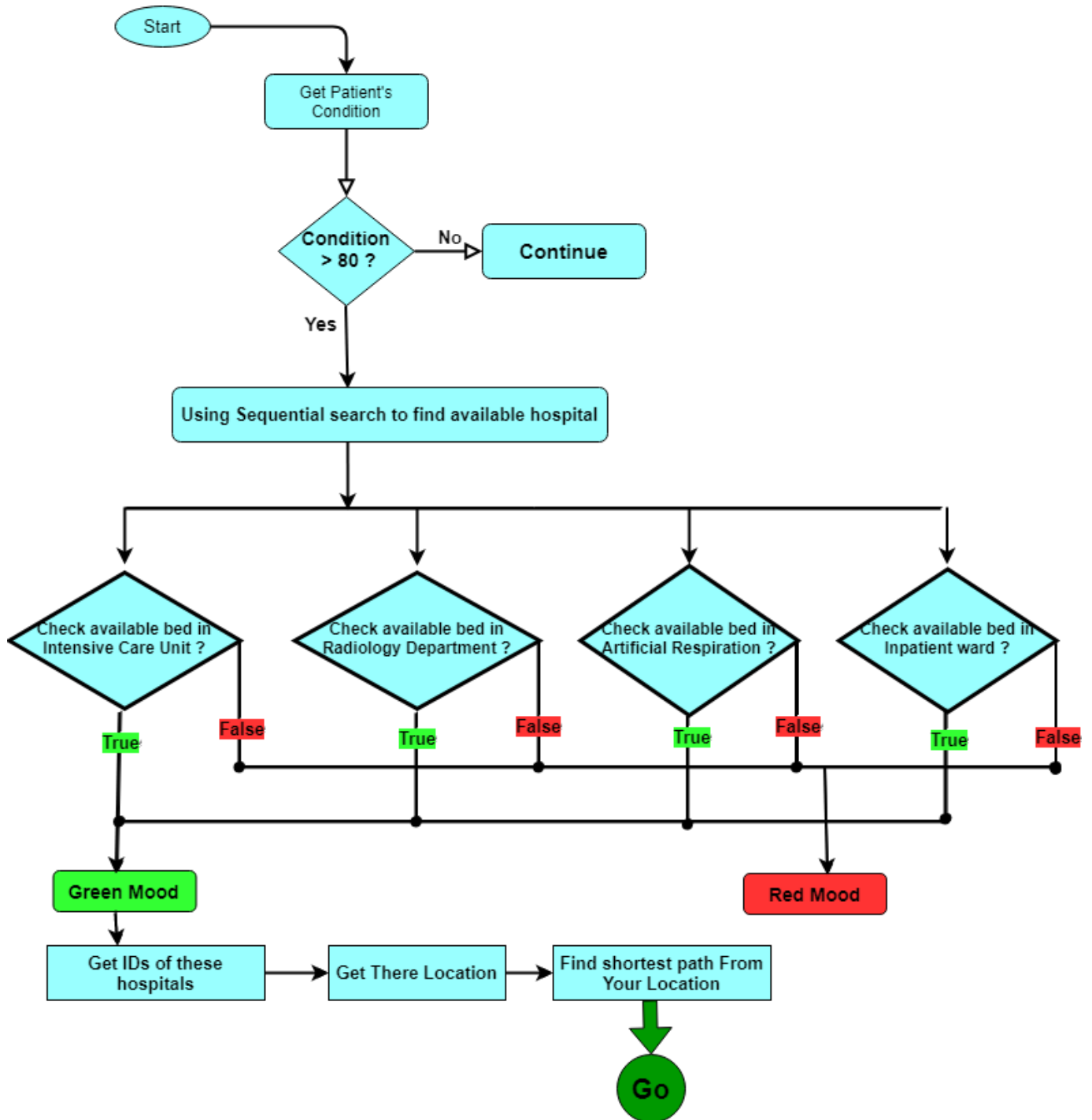
- 1- The patient can view information about hospitals across the city.
- 2- The information available in the application must be dynamic and in real-time, which means that the patient can see the available and current information about the hospitals at anytime and anywhere across the city.

- 3- can view each hospital's exact location, bed capacity and the available medical resources to suit a patient's case.
- 4- can view inpatient bed capacity in real-time for the hospitals.
- 5- can find the nearest suitable hospital location for a patient's case with the right medical services for this patient's emergency situation, to guarantee a transfer in a speedy manner patient and, at the same time, provide the right health care at the right time in a speedy manner.

four criteria are used to decide which hospital is available:

- 1- Available bed in radiology department.
- 2- Available bed in intensive care unit.
- 3- Available bed for inpatient.
- 4- Available bed in artificial respiration.

Workflow Design

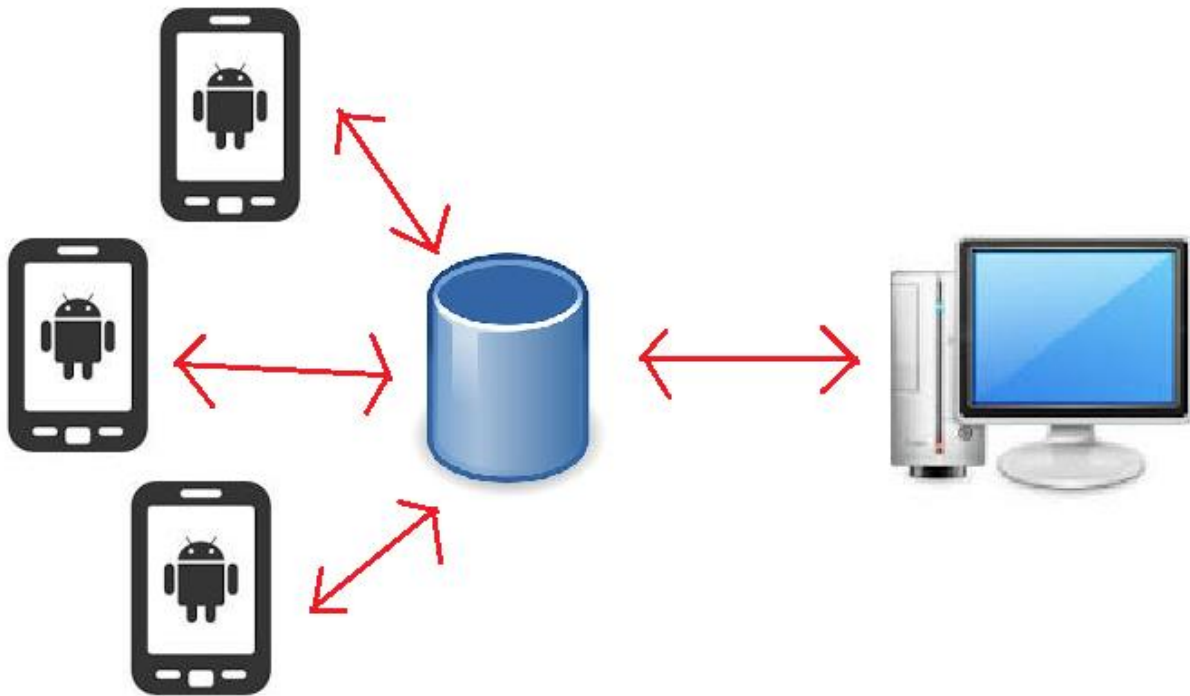


Steps:

1. Get patient's condition from machine learning model.
2. Check if patient's condition is critical (greater than 80 %).
 - a. If No Continue with home quarantine.
 - b. If Yes the algorithm will select all hospital within the region of patient, and using sequential search will find the available hospital based on the previous four criteria.
3. After get the list with available hospital will using shortest path first to find the nearest hospital from the patient.

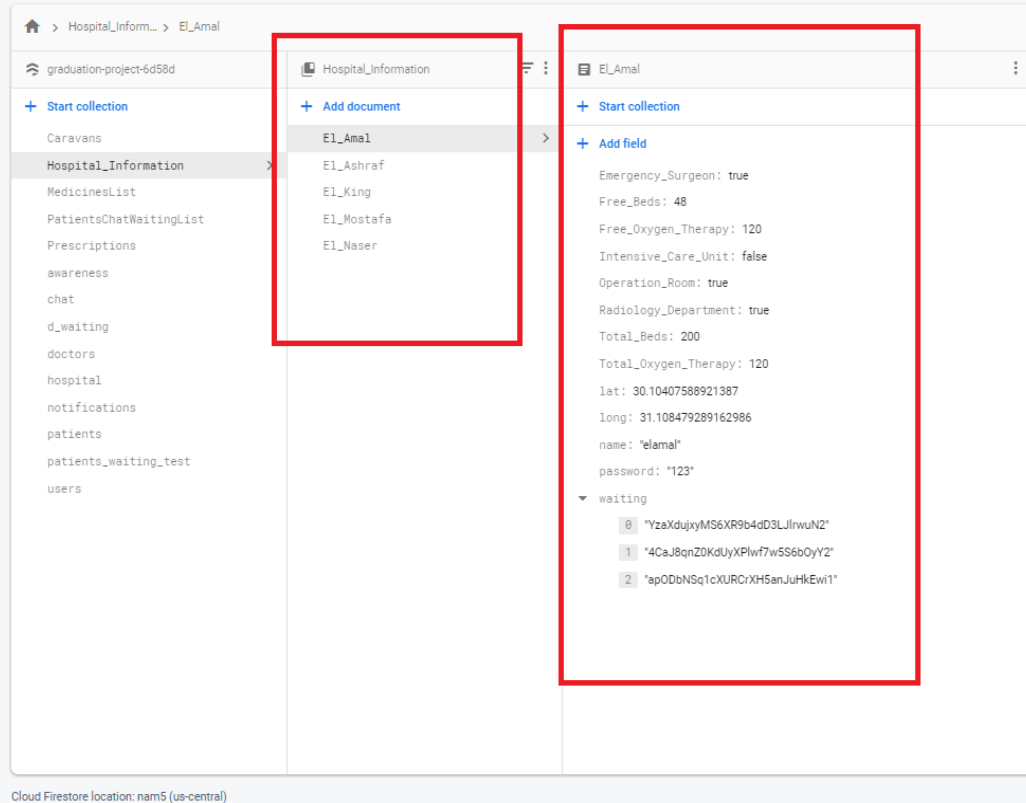
6 Chapter 6: Desktop application

6.1 Why was the Desktop Application built?



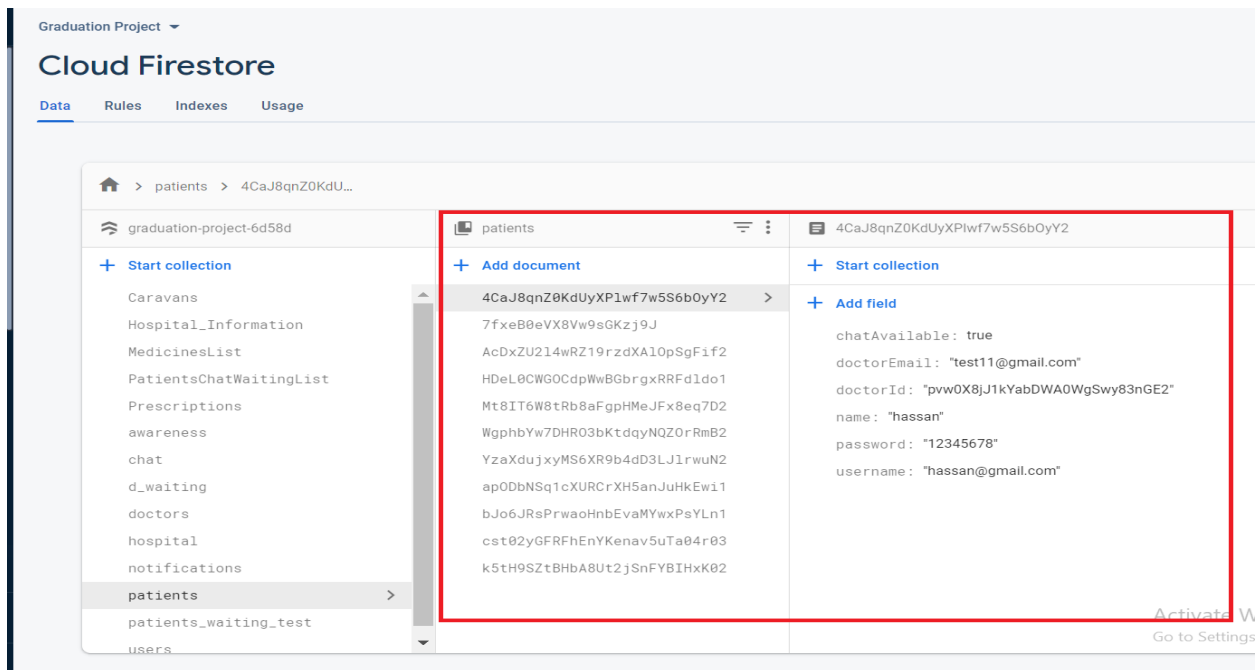
- When Mobile Application wants to direct a patient to a hospital, it needs information from this hospital.
- Therefore, The Desktop application was built and will be in every hospital, and it can add this information in Database.
- The Desktop Application must also be able to fetch information from the Database so that it can record new cases in the hospital.

6.2 Database

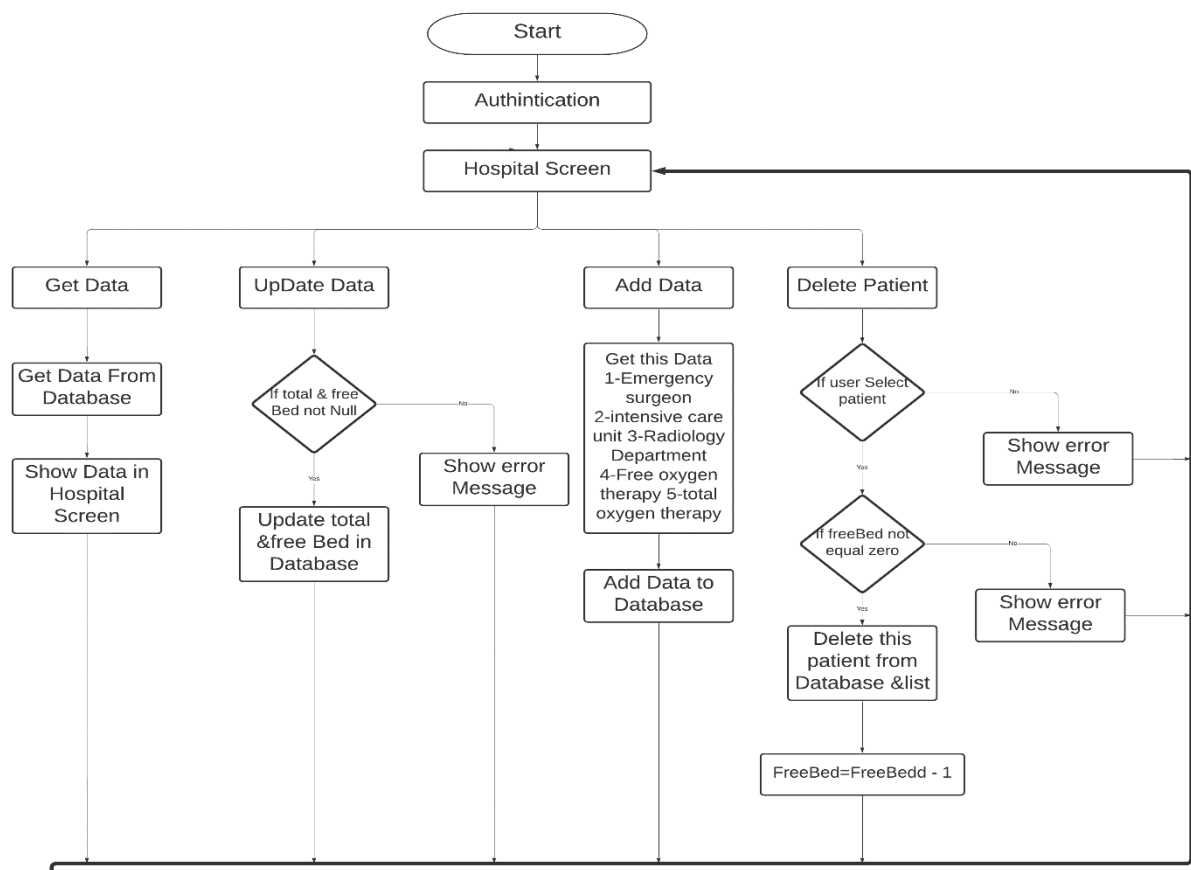


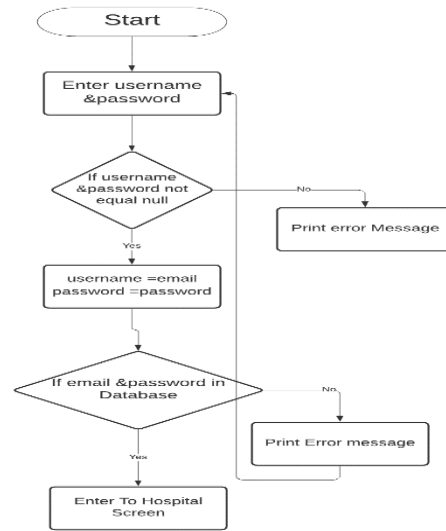
These are the databases that the desktop application is allowed to interact with, read from and update in this data.

The waiting list is an array in the Database containing the names of patients who need to be admitted to this hospital.

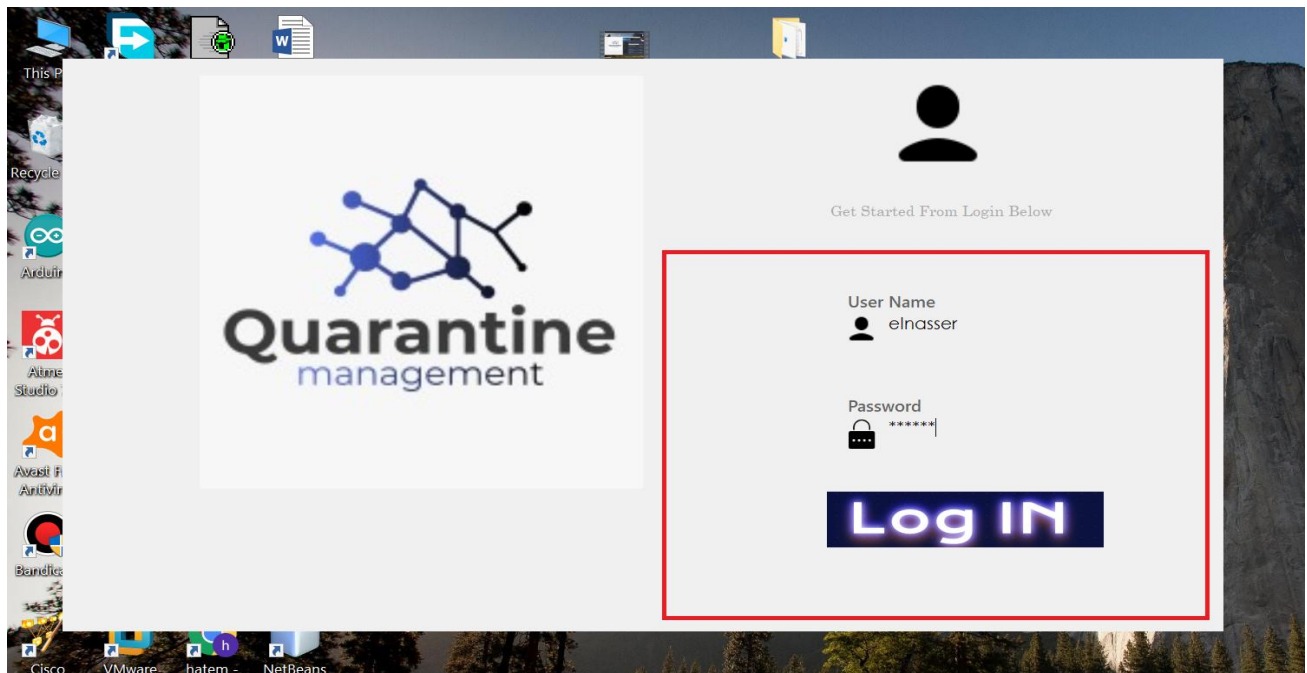


6.2 Flow Chart





6.3 Login



The first step that the user of this application takes in the hospital is to log in using the email and password given to that hospital.

6.4 Get information about the hospital

The screenshot shows a web application window titled "Show_patient". At the top, the username "elamal" is displayed in a red box. The main content area is divided into two sections. The left section, also outlined in red, contains a form for hospital information. It has two input fields: "Total Beds" with the value "200" and "Free Beds" with the value "48". To the right of these fields is a yellow "UPDate" button. Below these are two more input fields: "Free Oxygen Therapy" with the value "120" and "Total Oxygen Therapy" with the value "120". At the bottom of this section are four dropdown menus for "Emergency Surgeon" (set to "True"), "Intensive Care Unit" (set to "false"), "Operation Room" (set to "True"), and "Radiology Department" (set to "True"). A yellow "Add Data" button is positioned below these dropdowns. The right section is titled "Waiting Patient List" and contains a table with three columns: "User Name", "Name", and "ID". The table lists three patients: "hatem@helwan.com" (Name: hatem, ID: 1), "sdfsdf@fsdf.com" (Name: hossam, ID: 2), and "hassan@gmail.com" (Name: hassan, ID: 3). Below the table are two buttons: a blue "Get" button and a yellow "Delete" button. A circular arrow icon is located between the two buttons. At the bottom right, there is a watermark for "Activate Windows" with the text "Go to Settings to activate Windows."

User Name	Name	ID
hatem@helwan.com	hatem	1
sdfsdf@fsdf.com	hossam	2
hassan@gmail.com	hassan	3

At the beginning of the application, you need to fetch the information in the Database about this hospital.

1. Total Beds
2. Free Beds
3. Free Oxygen Therapy
4. Total Oxygen Therapy

5. Emergency surgeon
6. Intensive care unit
7. Operation Room
8. Radiology Department

6.5 UPdate and Add Data button

Show_patient

elamal

Total Beds

200

Free Beds

48

UPDATE

Free Oxygen Therapy

120

Total Oxygen Therapy

120

Emergency Surgeon

Intensive Care Unit

Operation Room

Radiology Department

True

false

True

True

Add Data

Waiting Patient List

User Name	Name	ID
hatem@helwan.com	hatem	1
sdfsdf@fsdf.com	hossam	2
hassan@gmail.com	hassan	3

Get

Delete

Activate Windows

Go to Settings to activate Windows.

When you click on the UPDate button, it will update this information Total Beds and Free Beds.

But when he clicks on Add Data, he will add this information or update it .

1. Free Oxygen Therapy
2. Total Oxygen Therapy
3. Emergency surgeon

4. Intensive care unit
5. Operation Room
6. Radiology Department

6.6 Get information about the patient

Show_patient

Total Beds

200

Free Beds

48

UPDATE

Free Oxygen Therapy

120

Total Oxygen Therapy

120

Emergency Surgeon

Intensive Care Unit

Operation Room

Radiology Department

True

false

True

True

Add Data

Waiting Patient List

User Name	Name	ID
hatem@helwan.com	hatem	1
sdfsdf@isdf.com	hossam	2
hassan@gmail.com	hassan	3

Get

Delete

Activate Windows

Go to Settings to activate Windows.

When the user presses the GET button, it will bring him the information of patients who need to enter this hospital.

1. User name
2. Name

6.7 Add patient To Hospital And it Delete From List

Show_patient

elamal

Total Beds200

Free Beds48

UPDATE

Free Oxygen Therapy120

Total Oxygen Therapy120

Emergency SurgeonTrueIntensive Care UnitfalseOperation RoomTrueRadiology DepartmentTrue

Add Data

Waiting Patient List

User Name	Name	ID
hatem@helwan.com	hatem	1
sdfsdf@fsdf.com	hossam	2
hassan@gmail.com	hassan	3

Get

Delete

The first thing here is for the user to choose from the list the name of the patient who needs to be admitted to the hospital, and then press the Delete button.

The application will delete this patient from the list and reduce the Free Beds by 1.

Chapter 7: Diagrams

Use case:

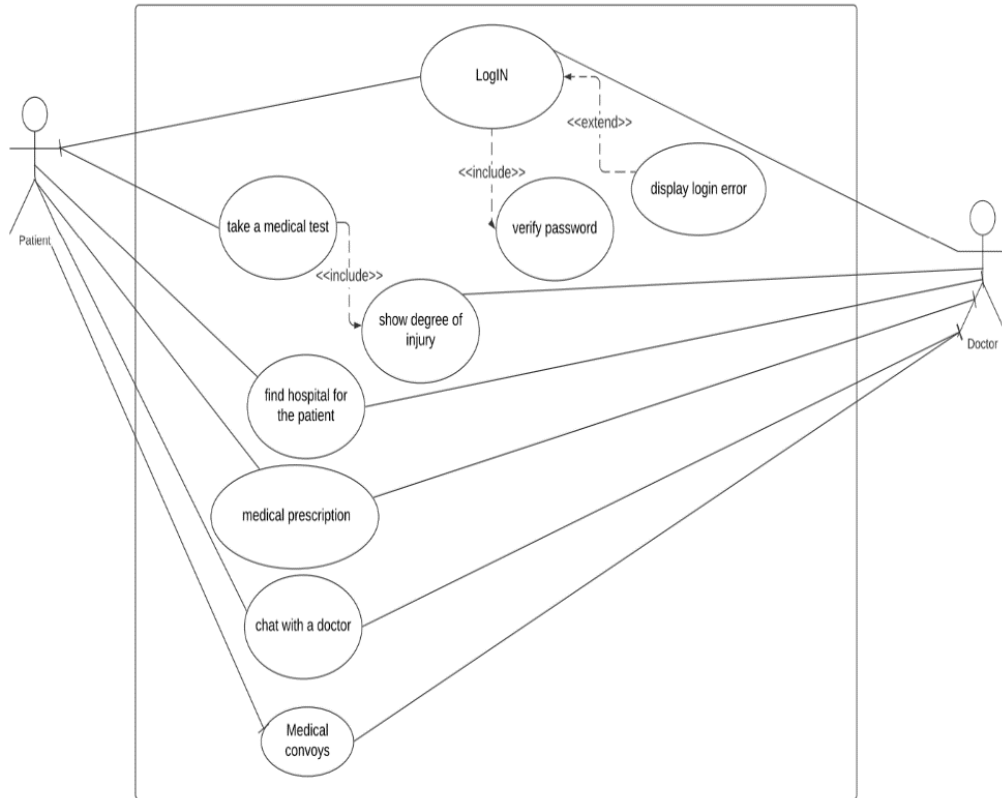


Figure 0-1 (use case)

The patient must login first then can make a medical test and show the result, he also can find hospital, chat with a doctor, receive a medical prescription or receive a medical convoys.

Sequence diagram:

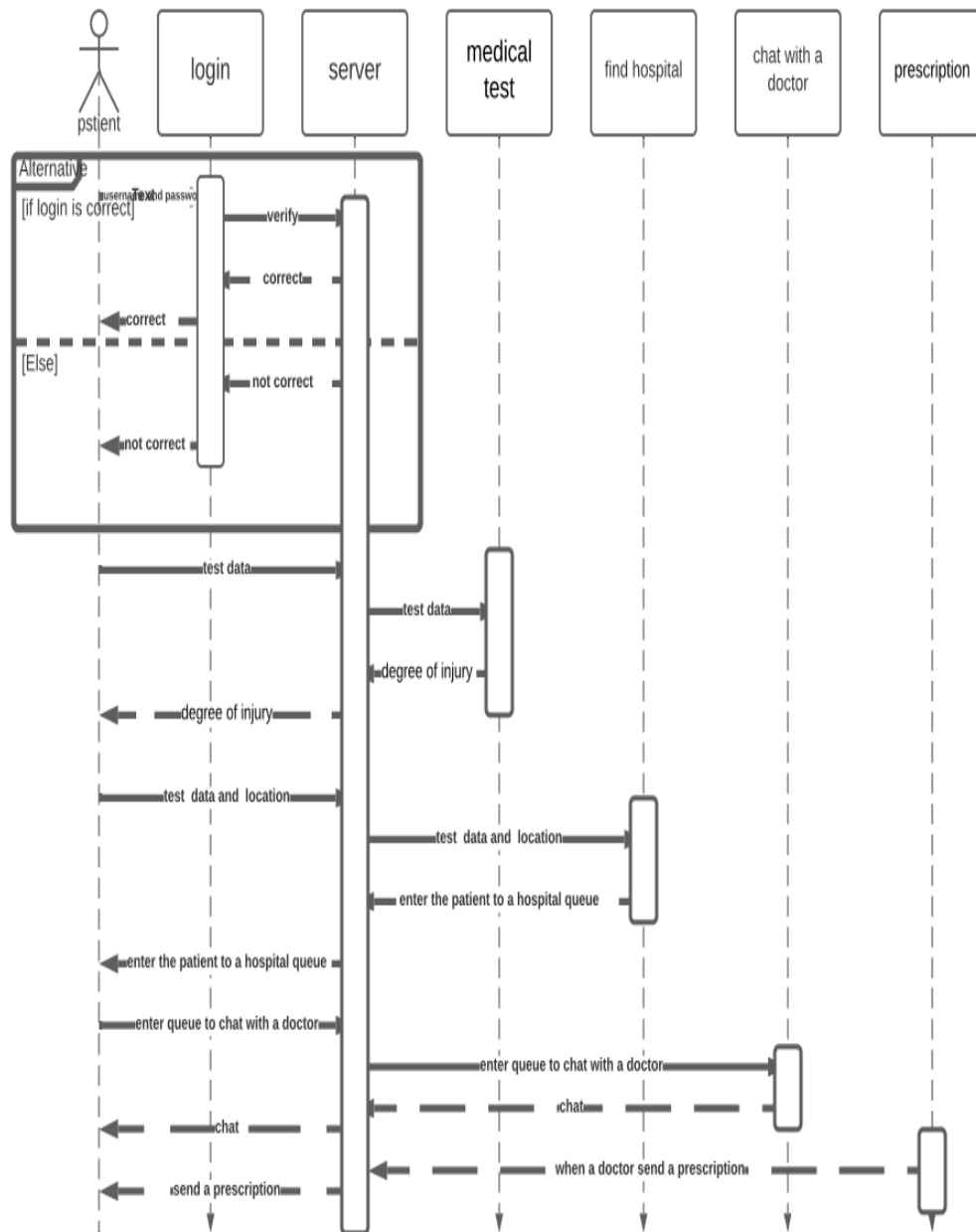


Figure 0-2 (sequence diagram)

Class diagram :

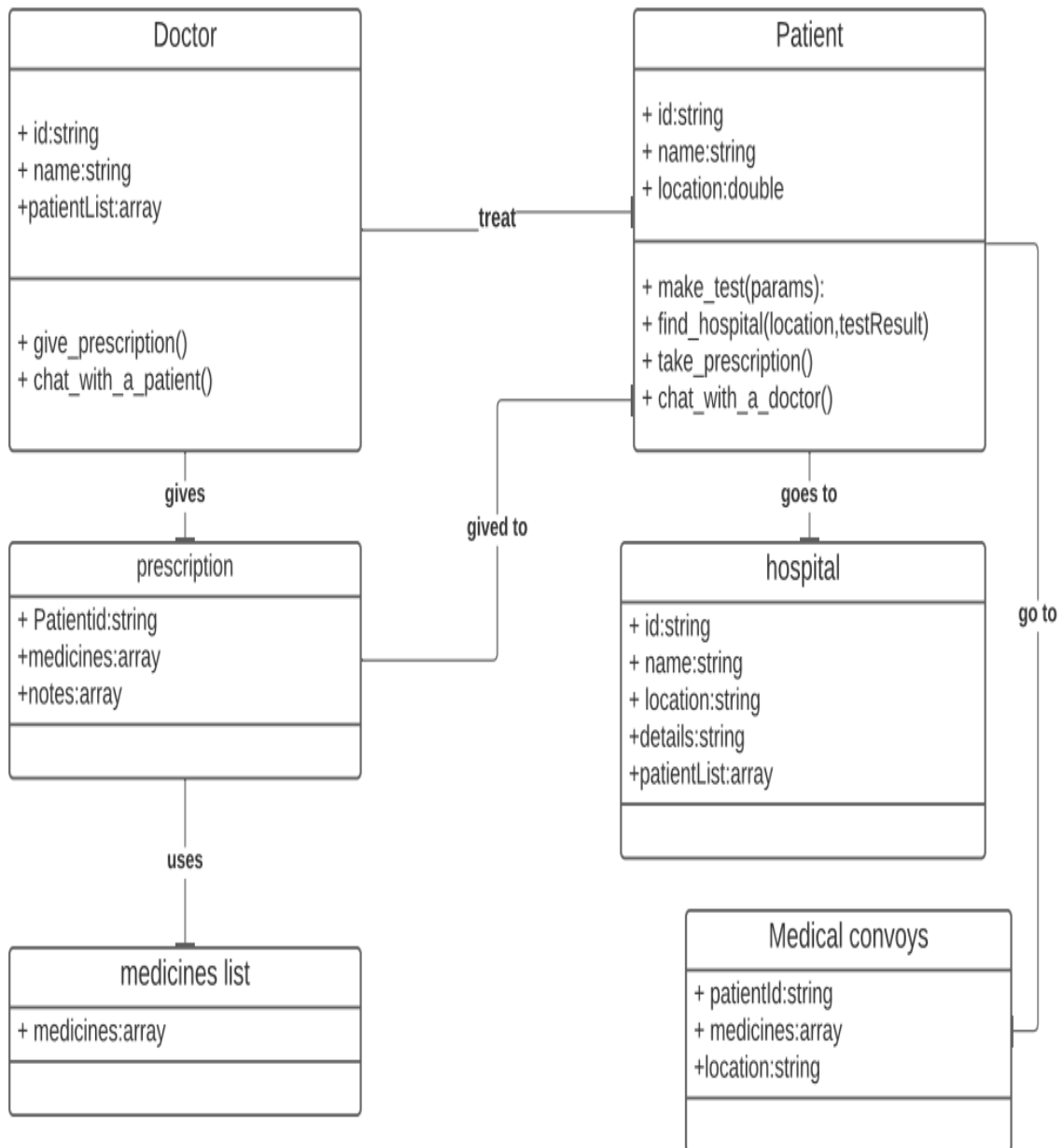


Figure 0-3 (class diagram)

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